

Moment and Prime-Trace Criteria for the Riemann Hypothesis, with Spectral Exclusions and Nonlocal Positivity

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Abstract

We develop moment, prime-trace, and continuous-kernel formulations of the Riemann hypothesis and establish spectral exclusions. For the normalized entire function $F(w) = \xi(\frac{1}{2} + i\sqrt{w})/\xi(\frac{1}{2})$, defined by its even power series, we prove the equivalence of RH, positivity of every reciprocal-square Hankel matrix generated by $-F'/F$, and a positive trace-class Fredholm determinant realization. The moment formulation is situated in classical Grommer–Hamburger theory, and an equivalent completed prime-trace identity is obtained in the absolutely convergent half-plane. The required counting law excludes ordinary compact elliptic spectra and regular boundary-power realizations. An exponential corridor matches both leading counting terms but has the wrong determinant; a controlled Whittaker expansion extends this exclusion to the full constant-coefficient Morse family with arbitrary scale, endpoint, constant shift, and fixed real separated boundary condition. Finite-to-infinite moment transfer and explicit trace and arithmetic error estimates delimit what numerical approximations establish. Exact Laplace and meromorphic identities connect the same target to Suzuki’s screw kernel and nonlocal characteristic ratio, retaining zero multiplicity and distinguishing the relevant convergence topologies. Within TCGS–SEQUENTION, conditional Gram-transfer statements specify the identities an independently defined source/readout realization must satisfy. The remaining step is a source-derived positive representation satisfying the complete arithmetic identity, or the required nonlocal analytic limit. The Riemann hypothesis is not proved.

Keywords: Riemann hypothesis; TCGS–SEQUENTION; 4D-Counterduction; Counterspace; Extrinsic Constitutive Law; source–shadow geometry; Hamburger moment problem; Fredholm determinant; spectral geometry; prime numbers; selector invariance; conservative cochains; boundary response; Schur complement; trace-norm convergence.

1 Introduction

The Riemann hypothesis concerns the zeros of a particular analytic function, not an unspecified resemblance between order and irregularity. Its arithmetic importance comes from the fact that

the Riemann zeta function simultaneously has a Dirichlet-series representation over the positive integers and an Euler-product representation over the primes. Riemann’s analytic continuation and explicit-formula programme connect these descriptions to the location of complex zeros [1, 2]. The hypothesis remains listed as unresolved by the Clay Mathematics Institute [3]. That status fixes the question to be addressed; it does not determine which mathematical constructions should be attempted.

TCGS–SEQUENTION supplies a different starting language from conventional arithmetic: a complete non-temporal four-dimensional source, a three-dimensional operational Shadow, a conserved singular structure, and an Extrinsic Constitutive Law (ECL) governing admissible readout. Its geometric and operator formulations distinguish a source configuration from its informational organization, the complete readability profile from one selected value, and an observable from the mathematical map used to represent it [4, 5, 6]. An application to number theory must retain these distinctions. In particular, neither an integer nor a complex zero is identified with a source singularity merely because the framework uses singular structure elsewhere.

The central research question is therefore constructive: can a source-grounded geometric representation produce a positive operator whose determinant or trace is *exactly* the relevant arithmetic function? Self-adjointness alone is insufficient. There are innumerable self-adjoint operators with spectra unrelated to prime numbers. Conversely, an operator defined retrospectively from zeros already assumed to be real cannot serve as an independent proof of their reality. The arithmetic identification and the positivity mechanism must be established without one being concealed inside the definition of the other. This is the central difficulty of spectral approaches, including the Hilbert–Pólya perspective and its semiclassical developments [7].

We pursue two complementary constructions. The first extracts a moment geometry directly from the completed zeta function. Its positive realization would settle the zero-location question. The second starts with an explicit positive geometric operator and asks whether its spectral determinant equals the arithmetic target. These directions meet at an exact trace identity. The first construction supplies arithmetic specificity but leaves universal positivity open; the second supplies positivity and the correct leading spectral density but fails arithmetic specificity. Their comparison identifies a concrete missing bridge rather than an unspecified demand for further formalization.

The moment approach belongs to an established mathematical landscape. Li’s criterion expresses the hypothesis through positivity of a different sequence [8]; Cardon and Roberts relate a Riemann-hypothesis condition, with simplicity, to orthogonal polynomials [9]; Voros develops zeta functions over zeta zeros [10]. The real-zero/Hankel method originates in Grommer’s criterion [11], whose inverse-power indexing is explicitly restated by Baricz and Štampach [27, Theorem 10]. Levin provides the general entire-function setting [26]; related Laguerre and Turán inequalities concern different positivity conditions [28]. Section 6.3 gives the precise even/odd block relation to the moments used here rather than identifying the differently indexed criteria by name alone. The moment equivalence below is not claimed as a new discovery. It is proved in the normalization needed for the present source-to-readout construction, with the dependence of every subsequent result made explicit.

The proved contribution is the resulting mathematical architecture and its executed tests. These proofs use classical analysis and explicit linear-algebraic hypotheses, not the TCGS axioms as additional mathematical premises. The framework supplies the realization question and its source/readout restrictions; Assumption 8.1 remains an unconstructed arithmetic identification. We obtain an ECL-to-arithmetic Gram-transfer theorem; a finite-rank obstruction; a prime-side trace identity sufficient and necessary for a positive determinant realization; a compact-elliptic spectral exclusion; and an explicit determinant mismatch for a calibrated exponential model. The

exponential construction is a rescaled member of the classical Bessel–Morse family, not a newly discovered solvable potential [16]. The substantive calculation is the comparison with the fully normalized Riemann function, including the term that prevents equality despite agreement of the leading counting law. The expanded comparison includes all fixed real separated endpoint conditions and constant spectral shifts; a controlled second-order coefficient closes the Robin case after its first-order matching succeeds. The leading counting term also fixes the product of the exponential rate and the square root of the kinetic coefficient, extending the exclusion to the full constant-coefficient Morse family with rate ratio 2 : 1.

This distinction also determines the role of the framework. Non-temporal Counterspace is the governing starting point, not a hypothesis tested by the numerical signs in this paper. The object exposed to mathematical failure is the *particular arithmetic realization*. A failed potential or an unproved Gram identity cannot be repaired by restating source completeness. Equally, rejecting one potential does not establish a no-go theorem for every realization compatible with the framework. The paper carries out the local construction, proves what it proves, and specifies the exact condition it does not obtain.

The conservative source-to-readout and reciprocal fractal-foam constructions provide a further route through compatible cochains, shared singular-boundary data, and interior elimination [20, 21]. We formulate its positive interface response without identifying it with the full bulk determinant, and connect finite refinements to arithmetic positivity through an all-degree limit theorem. Quantitative trace and prime-tail estimates then give certified rejection conditions. These additions execute the algebraic reduction and limit implications, while leaving the source-derived arithmetic identification explicit.

Recent arithmetic spectral constructions sharpen the intermediate representation problem. The extremal real-zero theorem of Connes and van Suijlekom, the arithmetic rank-one construction of Connes–Consani–Moscovici, and Suzuki’s screw-function formulation lead to different operators and limiting targets [30, 31, 29, 33]. Appendix E gives the exact normalization links, an all-window constitutive Gram-transfer implication, and finite-window error estimates. It also distinguishes the meromorphic characteristic ratio from the determinant F , preserving multiplicity information rather than comparing zero sets alone.

2 Source geometry and the arithmetic realization problem

2.1 The governing architecture

The four governing requirements are stated in their source–shadow order. Whole Content requires a complete non-temporal four-dimensional Counterspace, represented geometrically by $(\mathcal{C}^4, G, \Psi)$. Identity of Source supplies a distinguished origin p_0 and a conserved singular set

$$\mathcal{S} = \text{Orb}(p_0). \quad (2.1)$$

Shadow realization supplies an admissible immersion $X : \Sigma^3 \hookrightarrow \mathcal{C}^4$ and pullback data $(g, \psi) = X^*(G, \Psi)$. Parsimony requires one source-grounded constitutive/readability architecture rather than independently adjustable source laws for each application [4].

The foliation, its label, and operational clock time have distinct roles. An admissible foliation $\mathcal{F}$ is geometric readout structure. Its ordering label λ_g is defined only up to admissible monotone reparameterization. Operational time τ_{op} is a functional of Shadow clocks, correlations, and records. Thus

$$\mathcal{F} \not\equiv \lambda_g \not\equiv \tau_{\text{op}}, \quad t_{\text{ont}} \notin \text{Dim}(\mathcal{C}^4). \quad (2.2)$$

Here $\text{Dim}(\mathcal{C}^4)$ denotes the catalogue of source dimensions rather than an additional mathematical operator. The notation prevents an analytical parameter introduced later from being mistaken for an ontic temporal direction [4].

2.2 Law, profile, readout, and observable

Let $\Xi \in \mathcal{X}_{\text{adm}}$ denote an admissible source configuration. Boldface distinguishes this source datum from the conventional Riemann function $\Xi(z)$. The typed informational map is

$$\text{Info} : \mathcal{X}_{\text{adm}} \longrightarrow \mathcal{I}_{\text{src}}, \quad M_{\Xi} = \text{Info}(\Xi). \quad (2.3)$$

For selectors $\sigma = (X, [s], \mathcal{B})$, let $\mathcal{Y}_{\sigma}$ be the corresponding readout spaces and $\mathcal{P}_{\text{read}} = \prod_{\sigma} \mathcal{Y}_{\sigma}$. The ECL and its evaluations are

$$\mathcal{E}_{\mathcal{S}} = \text{Read}_{\mathcal{S}, G} \circ \text{Info}, \quad \mathcal{E}_{\mathcal{S}} : \mathcal{X}_{\text{adm}} \rightarrow \mathcal{P}_{\text{read}}, \quad (2.4)$$

$$\mathcal{E}_{\mathcal{S}, \Xi} = \mathcal{E}_{\mathcal{S}}(\Xi), \quad \mathcal{E}_{\mathcal{S}, \Xi}^{\sigma} = \text{ev}_{\sigma} \mathcal{E}_{\mathcal{S}, \Xi}. \quad (2.5)$$

An independently specified physical-readout map $\pi_{\sigma}^{\text{phys}}$ then gives

$$\mathcal{O}_{\mathcal{S}}^{\sigma} = \pi_{\sigma}^{\text{phys}} \circ \text{ev}_{\sigma} \circ \text{Read}_{\mathcal{S}, G} \circ \text{Info}. \quad (2.6)$$

This composition is the relevant 4D-Counterduction relation. It is not Counterspace itself, a second source manifold, or an independently evolving source process [4, 6].

In particular, the map $\mathcal{E}_{\mathcal{S}}$, its profile value, and a scalar function subsequently used in an arithmetic calculation are not interchangeable. The scalar function F introduced below is an analytic arithmetic target. No equation $F = \mathcal{E}_{\mathcal{S}}$ is asserted. Such an equality would erase both the source dependence of the ECL and the domain and codomain of the arithmetic function. An arithmetic application must instead specify a representation connecting an admissible source reduction to F , while retaining the distinction between the mathematical representation and the source it represents [4].

2.3 The legitimate positive operator

The relevant operator construction is based on a physical-readout derivative, not an untyped Hessian of a profile-valued law. At a reference source configuration Ξ_0 , suppose

$$L_{\sigma, 0} = D\mathcal{O}_{\mathcal{S}}^{\sigma}[\Xi_0] \quad (2.7)$$

is densely defined and closable between specified Hilbert completions. Writing L_{σ} for its closure, the natural operator

$$O_{\sigma} = L_{\sigma}^{*} L_{\sigma}, \quad \text{Dom}(O_{\sigma}) = \{v \in \text{Dom}(L_{\sigma}) : L_{\sigma} v \in \text{Dom}(L_{\sigma}^{*})\} \quad (2.8)$$

is positive and self-adjoint. Its quadratic form is

$$\langle v, O_{\sigma} v \rangle = \|L_{\sigma} v\|^2. \quad (2.9)$$

The mathematical statement is standard closed-operator theory [14]; its use here follows the explicitly typed observability construction in the framework [6].

The interpretation matters. Equation (2.9) measures squared readout sensitivity in the declared norm. It does not automatically measure energy, nor does it imply that O_{σ} has a discrete spectrum, a trace-class inverse, or the zeta zeros as eigenvalues. Banach charts, Hilbert metrics, derivative domains, closability, and selector comparison maps are additional analytic realization data. They are not inferred from the number of source dimensions [6].

2.4 What arithmetic must add

The formal inputs $(\mathcal{C}, G, \Psi, p_0, \mathcal{S})$ do not yet specify an action of integer multiplication, an Euler product, a prime-weighted trace, or a polynomial Gram map. Introducing such a map is legitimate as a proposed mathematical realization, but its construction must be displayed. A finite-dimensional protected response sector can be useful for a restricted physical application without encoding an infinite arithmetic spectrum. Section 8 makes this difference a rank obstruction rather than a verbal caution.

Nor is a one-dimensional analytical coordinate prohibited by a four-dimensional source ontology. A coordinate on an auxiliary function space, a reduced corridor, or a Mellin transform is not a claim that the physical source has become one-dimensional. The distinction follows the framework's separation of source dimension, representation dimension, and operational carrier [4, 5]. It permits the half-line model below while preventing its formal success from identifying that half-line with Counterspace.

2.5 Conservative source cartography and reciprocal interfaces

Two geometric constructions give additional content to the realization problem: the conservative discrete source-to-readout architecture and the reciprocal fractal-foam companion [20, 21]. They supply distinct ingredients. The former specifies orientation, dual volumes, shared-flux cancellation, compatible cochains, boundary completion, and selector transport. The latter supplies a candidate phase field ϕ , reciprocal closed two-form, nonlinear potential U , and source-determined interface and corridor weights. Neither construction currently supplies an arithmetic insertion map or a prime trace. Their use below is restricted to those structures they actually define.

The conservative paper distinguishes the ontic ECL from its scientific representation by a hat,

$$\widehat{\mathcal{E}}_{\mathcal{S},G} = \text{Read}_{\mathcal{S},G} \circ \text{Info}. \quad (2.10)$$

The symbol $\mathcal{E}_{\mathcal{S}}$ in Equation (2.4) is used at this same formal-map level throughout the present analysis. No finite operator, equation, or mesh is identified exhaustively with the ontic law. This clarification preserves the map/profile/readout types while adopting the explicit distinction in the two geometric papers [20, 21].

Let r index a declared refinement sequence on regularized source domains

$$\Omega_{\varrho_r} = \mathcal{C} \setminus N_{\varrho_r}(\mathcal{S}), \quad h_r \downarrow 0, \quad \varrho_r \downarrow 0. \quad (2.11)$$

The sequence may additionally resolve diffuse interfaces or expand a noncompact domain; every such limit and its order must be specified. At fixed r , conservative assembly requires a shared flux with $\widehat{Q}_{kj} = -\widehat{Q}_{jk}$. A reciprocal-form realization additionally requires

$$d_{q+1,r}d_{q,r} = 0, \quad d_r X_{\sigma,r}^* = X_{\sigma,r}^* d_r. \quad (2.12)$$

The nodal volumes and directed hyperareas do not supply the full cochain complex or its Hodge matrices. They cannot be used as though the reciprocal two-form were an ordinary scalar nodal variable [20, 21]. Stable approximation of the associated Hilbert complex requires suitable subcomplexes and bounded commuting projections in addition to the algebraic identities [22].

Gate T of the geometric constructions fixes the source action, singular orbit, complement topology, periods, and induced carrier classes before a harmonic sector is admitted. For example, the isolated-point complement in a four-ball has trivial degree-two cohomology. Gate S then requires

all singular-boundary channels to descend from one declared functional $\mathcal{B}_S$, with coefficients and excision scaling fixed independently of the target. Its prescribed variation jointly generates the phase flux, nonlinear-potential flux, reciprocal current, and traction:

$$\delta\mathcal{B}_S = \int_{\partial N_{\mathcal{G}_r}(S)} \left[-Q_S^\phi \delta\phi - Q_S^U \delta U + \langle J_S, \delta A_\parallel \rangle - \langle t_S, \delta X \rangle \right] dA_G. \quad (2.13)$$

The geometric papers state this requirement but do not construct the required shared functional. It therefore enters the response construction as explicit missing model data, not as a completed source derivation [20, 21].

The companion's corridor metric is $\tilde{G} = w_{\text{cor}}^{-2} G$, where w_{cor} denotes its source-determined readability weight, distinguished here from the complex arithmetic variable w . Its multifractal weight spectrum is not an eigenvalue counting theorem. Neither a corridor nor a harmonic period may be assigned a prime label and then cited as a derivation of that label. The reported median-dual and manufactured-curvature checks are implementation evidence within their stated tests; they do not establish spectral convergence, an arithmetic determinant, or the unresolved boundary functional [20, 21].

3 The arithmetic target and its entire normalization

For $\Re s > 1$, define

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_{p \text{ prime}} (1 - p^{-s})^{-1}. \quad (3.1)$$

The domain in Equation (3.1) is essential: its absolutely convergent Euler product is not used on the critical line. Analytic continuation gives the meromorphic zeta function, and the completion

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s) \quad (3.2)$$

is entire of order one. It satisfies $\xi(s) = \xi(1-s)$ and $\xi(\bar{s}) = \overline{\xi(s)}$. Its zeros are precisely the nontrivial zeta zeros, with multiplicity [1, 2, 17].

Set

$$\Xi(z) = \xi\left(\frac{1}{2} + iz\right), \quad F(w) = \frac{\Xi(\sqrt{w})}{\Xi(0)}. \quad (3.3)$$

The second expression is notation for an entire power series, not a branch choice. Since Ξ is even,

$$\Xi(z) = \sum_{j=0}^{\infty} c_j z^{2j}, \quad F(w) = \frac{1}{c_0} \sum_{j=0}^{\infty} c_j w^j. \quad (3.4)$$

Consequently F is a real entire function of order $1/2$ and $F(0) = 1$. If $\rho = \beta + i\gamma$ is a zero with $\gamma > 0$, its corresponding zero of F is

$$w_\rho = -(\rho - \tfrac{1}{2})^2 = (\gamma + i(\tfrac{1}{2} - \beta))^2. \quad (3.5)$$

Exactly one representative of the pair $\rho, 1 - \rho$ is counted by taking $\gamma > 0$. Off-line zeros give conjugate nonreal values of w_ρ . On-line zeros give $w_\rho = \gamma^2 > 0$.

Lemma 3.1 (Genus-zero normalization). *Enumerate the zeros of F as (w_j) with multiplicity. Then*

$$\sum_j |w_j|^{-1} < \infty, \quad F(w) = \prod_j \left(1 - \frac{w}{w_j}\right). \quad (3.6)$$

Moreover $F(-x) > 0$ for $x \geq 0$.

Proof. The standard zero-counting estimate $N(T) = O(T \log T)$, where $N(T)$ counts nontrivial zeros with $0 < \Im \rho \leq T$, implies $\sum_{\Im \rho > 0} |\rho - 1/2|^{-2} < \infty$ [2]. This is the first statement. Since F has order $1/2 < 1$, its Hadamard factorization contains no nonconstant exponential factor. The normalization $F(0) = 1$ gives Equation (3.6). Finally, Section 5 derives the positive-kernel formula

$$\xi(\tfrac{1}{2} + r) = \int_0^\infty \Phi(u) \cosh(ru) du > 0 \quad (r \in \mathbb{R}).$$

Taking $r = \sqrt{x}$ and using the functional equation proves $F(-x) > 0$. □

Corollary 3.2. *The Riemann hypothesis is equivalent to every zero of F being positive real.*

Proof. Under the hypothesis Equation (3.5) gives $w_\rho = \gamma^2$. Conversely, if $w_\rho > 0$, the number $z = \gamma + i(1/2 - \beta)$ has positive real part and real positive square. Its imaginary part is therefore zero, giving $\beta = 1/2$. □

This normalization removes an avoidable ambiguity in spectral programmes. A self-adjoint operator with eigenvalues γ is usually discussed on both sides of zero. A positive trace-class operator will instead have eigenvalues γ^{-2} . These are different spectral variables, with different counting laws and operator classes. Keeping the square and inverse explicit prevents an asymptotic calculation in one variable from being mistaken for a result in another.

4 Why the critical half is natural but not sufficient

4.1 Reflection does not force a zero onto its fixed set

The holomorphic map $s \mapsto 1 - s$ has the single fixed point $s = 1/2$. The antiholomorphic reflection $s \mapsto 1 - \bar{s}$ has the critical line as its fixed set. The functional equation and complex conjugation together preserve the zero set under these maps, but invariance of a set under a reflection does not imply that each of its points is fixed.

For example, for $a, b > 0$,

$$P(z) = ((z - a)^2 + b^2)((z + a)^2 + b^2) \quad (4.1)$$

is even and real on the real axis, while its zeros are $\pm a \pm ib$. Taking $0 < b < 1/2$ places the corresponding $s = 1/2 + iz$ zeros inside the critical strip but off its central line. This elementary example is not a counterexample to RH: it has no zeta Euler product. It refutes only the inference from reflection symmetry to pointwise zero confinement.

4.2 The half-density calculation

There is nevertheless a precise analytical reason for a factor $1/2$ in a scale representation. On $L^2(\mathbb{R}_+, dx)$ define, for $a > 0$,

$$(D_a f)(x) = a^{1/2} f(ax). \quad (4.2)$$

A change of variables gives $\|D_a f\|_2 = \|f\|_2$. Under the logarithmic unitary map

$$(Uf)(r) = e^{r/2} f(e^r), \quad U : L^2(\mathbb{R}_+, dx) \longrightarrow L^2(\mathbb{R}, dr), \quad (4.3)$$

dilation becomes translation: $UD_{e^v}U^{-1}g(r) = g(r+v)$.

Proposition 4.1 (Half-density generator). *The operator*

$$B = -i \left(x \frac{d}{dx} + \frac{1}{2} \right) \quad (4.4)$$

on $C_c^\infty(\mathbb{R}_+)$ is essentially self-adjoint. Its closure is unitarily equivalent to $-i d/dr$ on $L^2(\mathbb{R})$ and has purely continuous spectrum $\mathbb{R}$.

Proof. Differentiating Equation (4.3) gives $UBU^{-1} = -i d/dr$ on $C_c^\infty(\mathbb{R})$. The Fourier transform realizes the closure of the latter as multiplication by a real variable, with domain consisting of functions whose Fourier transforms remain square-integrable after multiplication by that variable. This multiplication operator is self-adjoint and has no square-integrable point eigenvectors. Its spectrum is $\mathbb{R}$. Unitary equivalence gives the assertion. $\square$

The Mellin transform of f on $s = 1/2 + it$ becomes a Fourier transform of Uf :

$$\int_0^\infty f(x) x^{-1/2+it} dx = \int_{-\infty}^\infty (Uf)(r) e^{itr} dr. \quad (4.5)$$

The real part $1/2$ is fixed here by the Lebesgue measure dx and its half-density correction. It is not obtained by subtracting three source dimensions from four or by treating a temporal coordinate as imaginary. The choice $L^2(\mathbb{R}_+, dx/x)$ shifts the corresponding normalization, illustrating why the measure must be declared before assigning geometric significance to the exponent.

This calculation gives a real spectral axis but not the arithmetic spectrum. The dilation operator has continuous spectrum, whereas F has an infinite discrete zero set. Discreteness, multiplicity, and prime-sensitive trace data remain to be supplied. The calculation is thus a useful local geometric ingredient and not a proof of RH. Its role parallels the distinction between a formal dilation Hamiltonian and a fully specified spectral realization in the Berry–Keating programme [7].

5 Positive theta geometry and the nonlinear moment step

5.1 An exact positive-kernel representation

Let

$$\vartheta_+(x) = \sum_{n=1}^\infty e^{-\pi n^2 x}, \quad h(u) = e^{u/2} \vartheta_+(e^{2u}), \quad u \geq 0. \quad (5.1)$$

The theta transformation underlying the functional equation gives the entire identity

$$\xi(s) = \frac{1}{2} + \frac{s(s-1)}{2} \int_1^\infty \vartheta_+(x) \left(x^{s/2} + x^{(1-s)/2} \right) \frac{dx}{x}. \quad (5.2)$$

This is a standard form of Riemann's theta representation [1, 2, 17]. With $s = 1/2 + iz$ and $x = e^{2u}$, it becomes

$$\Xi(z) = \frac{1}{2} - 2(z^2 + \frac{1}{4}) \int_0^\infty h(u) \cos(zu) du. \quad (5.3)$$

The full theta relation $\vartheta(x) = x^{-1/2}\vartheta(1/x)$, where $\vartheta = 1 + 2\vartheta_+$, implies $\vartheta'(1) = -\vartheta(1)/4$. Therefore

$$h'(0) = \frac{1}{2}\vartheta_+(1) + 2\vartheta'_+(1) = -\frac{1}{4}. \quad (5.4)$$

Twice integrating by parts in Equation (5.3), with all terms at infinity vanishing, yields

$$\Xi(z) = \int_0^\infty \Phi(u) \cos(zu) du, \quad \Phi(u) = 2(h''(u) - \frac{1}{4}h(u)). \quad (5.5)$$

Termwise differentiation gives the explicit formula

$$\Phi(u) = 4 \sum_{n=1}^\infty \left(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2} \right) e^{-\pi n^2 e^{2u}}. \quad (5.6)$$

Every summand is strictly positive for $u \geq 0$, because $2\pi n^2 e^{2u} > 3$. The double-exponential decay justifies differentiation and integration on compact subsets of the complex z -plane. In particular, Equation (5.5) establishes the real-axis positivity used in Lemma 3.1.

5.2 Raw moments and logarithmic moments are different objects

Define

$$A_{2j} = \int_0^\infty \Phi(u) u^{2j} du, \quad F(w) = \sum_{j=0}^\infty f_j w^j, \quad f_j = \frac{(-1)^j A_{2j}}{(2j)! A_0}. \quad (5.7)$$

The A_{2j} are moments of an explicitly positive measure. Their ordinary moment matrices are positive. The arithmetic matrices needed for RH, however, are generated by a logarithmic derivative:

$$M(w) = -\frac{F'(w)}{F(w)} = \sum_{n=0}^\infty m_n w^n \quad \text{near } w = 0. \quad (5.8)$$

Equating coefficients in $MF = -F'$ gives

$$m_n = -(n+1)f_{n+1} - \sum_{k=1}^n f_k m_{n-k}. \quad (5.9)$$

For example,

$$m_0 = -f_1, \quad m_1 = f_1^2 - 2f_2, \quad (5.10)$$

$$m_2 = -f_1^3 + 3f_1 f_2 - 3f_3. \quad (5.11)$$

This nonlinear transformation is exactly where positivity ceases to be automatic. The positivity of Φ supplies a Hilbert norm for raw polynomial moments in u , but it does not yet supply a Hilbert norm for the m_n . An ECL interpretation must address the latter norm, not substitute the former.

5.3 Why positivity of a Fourier kernel is insufficient

Let g be the centered Gaussian probability density whose Fourier transform is e^{-z^2} . The strictly positive even function

$$k(u) = 2g(u) + \frac{1}{2}g(u-1) + \frac{1}{2}g(u+1) \quad (5.12)$$

has entire Fourier transform

$$\widehat{k}(z) = e^{-z^2}(2 + \cos z). \quad (5.13)$$

Its zeros include $z = \pi \pm i \operatorname{arcosh} 2$, which are nonreal. Thus an even positive kernel and an entire transform do not suffice for real-rootedness. This example does not have the exact order or arithmetic structure of Ξ , and it is not offered as a counterexample inside the zeta class. It isolates the invalid positivity implication.

The geometric task is consequently sharper than producing a positive density. One must show that the *logarithmic* moment form induced by the actual theta kernel is positive in every polynomial degree. The following section makes that statement equivalent to the original hypothesis, without assuming the zeros are simple or computing their ordinates.

6 The positive moment criterion

6.1 An arithmetic quadratic form

For $N \geq 0$, define the real symmetric Hankel matrix

$$H_N = (m_{i+j})_{i,j=0}^N. \quad (6.1)$$

For a polynomial $p(t) = \sum_{j=0}^N c_j t^j$, set

$$q_\xi(p) = \sum_{i,j=0}^N \bar{c}_i c_j m_{i+j}. \quad (6.2)$$

The subscript records that the form is extracted from the specific Riemann completion, not selected from an arbitrary positive model. We use inner products conjugate-linear in the first variable throughout. For $c = a + ib$ with real vectors a, b , symmetry gives $c^* H_N c = a^T H_N a + b^T H_N b$; thus positivity over real coefficients is equivalent to positivity over complex coefficients.

The canonical product gives an exact arithmetic interpretation of its coefficients:

$$M(w) = \sum_j \frac{1}{w_j - w}, \quad m_n = \sum_j w_j^{-n-1}. \quad (6.3)$$

These series converge absolutely near the origin, and the first series converges normally on compact sets avoiding the zeros. Conjugate pairing makes every m_n real even without RH. With a sum over *all* nontrivial zeros, rather than the upper-half-plane representatives used here, the corresponding reciprocal-square power sums are $2m_n$ [10].

Lemma 6.1 (Analytic moment growth forces compact support). *Suppose $(m_n)_{n \geq 0}$ is a real Hamburger moment sequence and $\sum_{n \geq 0} m_n w^n$ has positive radius of convergence. Then every positive representing measure has compact support.*

Proof. Choose $r > 0$ inside the radius of convergence. Cauchy's estimate supplies a constant C_r with $|m_n| \leq C_r r^{-n}$. Let μ be a positive representing measure. If $\mu(\{|t| \geq A\}) = c > 0$ for some $A > r^{-1}$, then

$$m_{2n} = \int_{\mathbb{R}} t^{2n} d\mu(t) \geq cA^{2n}.$$

This contradicts $m_{2n} \leq C_r r^{-2n}$ as $n \rightarrow \infty$. Hence $\text{supp } \mu \subseteq [-r^{-1}, r^{-1}]$. $\square$

Theorem 6.2 (Reciprocal-square Hamburger criterion). *For the function F in Equation (3.3), the following are equivalent:*

1. *the Riemann hypothesis;*
2. *H_N is positive semidefinite for every $N \geq 0$;*
3. *H_N is positive definite for every $N \geq 0$;*
4. *$\det H_N > 0$ for every $N \geq 0$.*

No simplicity assumption on the zeros is required.

Proof. Assume RH. Write the positive ordinates, with multiplicity, as γ_j and put $\lambda_j = \gamma_j^{-2}$. Lemma 3.1 gives $\sum_j \lambda_j < \infty$. Define the finite positive measure

$$\mu_\xi = \sum_j \lambda_j \delta_{\lambda_j}. \quad (6.4)$$

Then $m_n = \int t^n d\mu_\xi(t)$, so

$$q_\xi(p) = \int |p(t)|^2 d\mu_\xi(t) = \sum_j \lambda_j |p(\lambda_j)|^2 \geq 0. \quad (6.5)$$

There are infinitely many distinct λ_j , since ξ has infinitely many zeros and each zero has finite multiplicity. A nonzero polynomial cannot vanish on all of them. Thus Equation (6.5) is strictly positive for nonzero p , proving statement 3.

Conversely, assume statement 2. The Hamburger moment theorem gives a positive Borel measure μ on $\mathbb{R}$ satisfying $m_n = \int t^n d\mu(t)$ [12]. Lemma 6.1 makes its support compact. Therefore

$$G_\mu(w) = \int_{\mathbb{R}} \frac{d\mu(t)}{1 - wt} \quad (6.6)$$

is holomorphic in each open nonreal half-plane and agrees with $M(w)$ near $w = 0$. Indeed, on a sufficiently small disc the geometric series can be integrated uniformly. The identity theorem for meromorphic functions then gives $G_\mu = -F'/F$ throughout each nonreal half-plane.

If F had a nonreal zero w_* of multiplicity $d \geq 1$, its logarithmic derivative would have the nonremovable local term

$$-\frac{F'(w)}{F(w)} = -\frac{d}{w - w_*} + \text{holomorphic part}.$$

This contradicts the holomorphy of Equation (6.6). Consequently all zeros of F are real. Lemma 3.1 excludes $(-\infty, 0]$, so all its zeros are positive. Corollary 3.2 proves RH.

Statement 3 implies statement 2. Statement 3 also implies statement 4. Conversely, statement 4 gives positivity of every leading principal minor of each finite H_N , and Sylvester's criterion gives statement 3. This completes the equivalence. $\square$

Theorem 6.2 is a reciprocal-square formulation within the classical real-zero/moment method [11, 12, 27]. The proof above supplies the analytic normalization and compact-support argument needed here; it does not claim priority for a new real-zero criterion. The relation to Grommer's inverse-power sequence is derived in Section 6.3. Its utility for TCGS-SEQUENTION is that the target is now a definite positive form, Equation (6.2), to which a readout norm can be compared.

Remark 6.3 (A distinction about minors). It is important that statement 4 uses *strictly positive* leading determinants. Merely requiring all leading determinants to be nonnegative is not, in general, a sufficient test for positive semidefiniteness of a real symmetric matrix. The unambiguous non-strict formulation is statement 2, which tests each complete matrix. No argument below substitutes one condition for the other.

6.2 What the first conditions actually test

The first two matrix conditions are

$$m_0 \geq 0, \quad m_0 m_2 - m_1^2 \geq 0. \quad (6.7)$$

Positivity of each individual m_n is weaker than the Hankel requirement. The second inequality already compares three logarithmic moments. In a positive realization it is a Cauchy-Schwarz inequality between the constant polynomial and the coordinate polynomial. In arithmetic form it is a nonlinear relation among the first three nonconstant coefficients of F .

This illustrates the exact conceptual gain. The question is not whether a list of scalar values looks positive, but whether all finite combinations define a compatible Hilbert geometry. Every degree enlarges the class of combinations that must be controlled. A positive measure for the raw theta moments does not automatically establish these logarithmic-moment inequalities, because Equation (5.9) does not preserve moment positivity by definition.

6.3 The classical Grommer criterion and the reciprocal-square indexing

Grommer's real-zero criterion is stated in terms of Hankel determinants of inverse powers of zeros [11]. We use the order-one formulation given explicitly in [27, Theorem 10]: for the class relevant here, the moments begin at the inverse *second* power, and strict positivity of the complete determinant sequence characterizes reality of all zeros. The Riemann function Ξ has real Taylor coefficients, order one, no zero at the origin, infinitely many distinct zeros, and the needed inverse-square summability. These hypotheses are important; strict positivity is not asserted for a finite-support zero set. General background on real entire functions is provided by Levin [26].

Let (z_ℓ) enumerate *all* zeros of Ξ , with multiplicity, and define

$$g_k = \sum_{\ell} z_{\ell}^{-k-2}, \quad G_n = (g_{i+j})_{i,j=0}^{n-1}, \quad n \geq 1. \quad (6.8)$$

Absolute convergence and the pairing $z \leftrightarrow -z$ give the exact identities

$$g_{2j} = 2m_j, \quad g_{2j+1} = 0. \quad (6.9)$$

Introduce the shifted Hankel matrix

$$H_N^{(+)} = (m_{i+j+1})_{i,j=0}^N. \quad (6.10)$$

Permuting the rows and columns of $\mathbf{G}_n$ into even and odd index order therefore yields

$$\mathbf{P}_n \mathbf{G}_n \mathbf{P}_n^* = 2H_{\lceil n/2 \rceil - 1} \oplus 2H_{\lfloor n/2 \rfloor - 1}^{(+)} \quad (6.11)$$

where an index -1 denotes an absent block. Equivalently, with the determinant of the empty matrix equal to one,

$$\det \mathbf{G}_n = 2^n \det H_{\lceil n/2 \rceil - 1} \det H_{\lfloor n/2 \rfloor - 1}^{(+)} \quad (6.12)$$

These formulas identify the normalization and the missing shifted block explicitly. The single tower in Theorem 6.2 is not literally the full Grommer matrix sequence. Its sufficiency uses additional structure of the fixed function F : genus-zero factorization, analyticity of the logarithmic moment transform, and the absence of zeros on $(-\infty, 0]$. The compact-support/pole argument proves positivity of its zeros from the unshifted tower; then Equation (6.4) gives both towers as positive moment matrices. For an arbitrary Hamburger sequence, unshifted positivity alone does not imply positivity of the shifted tower. For a Hamburger sequence, the additional condition $H_N^{(+)} \succeq 0$ for all N is precisely the Stieltjes support condition for a representing measure on $[0, \infty)$ [12]. Thus, for this particular F , Theorem 6.2 upgrades Hamburger positivity to Stieltjes positivity because Lemma 3.1 has already excluded $(-\infty, 0]$ from the zero set. No simplicity assumption is introduced by this comparison. The related Laguerre and Turán inequalities [28] are not substituted for this all-degree Hankel requirement.

7 Fredholm determinants and a prime-side closure identity

7.1 Positive trace-class realization

For a positive trace-class operator K on a separable Hilbert space, the Fredholm determinant is

$$D_K(w) = \det(\mathbf{I} - wK) = \prod_j (1 - w\lambda_j), \quad (7.1)$$

where (λ_j) are the positive eigenvalues with multiplicity. A kernel of K contributes no factor. The product converges locally uniformly because $\sum_j \lambda_j < \infty$, and its zeros are positive real [13].

Theorem 7.1 (Determinant and trace equivalences). *The following statements are equivalent:*

1. *RH holds;*
2. *there exists a positive trace-class operator K such that $F(w) = D_K(w)$ for every $w \in \mathbb{C}$;*
3. *there exists a positive trace-class operator K such that*

$$\mathrm{Tr} K^{n+1} = m_n \quad (n \geq 0). \quad (7.2)$$

Proof. Under RH, take K on ℓ^2 to be diagonal with eigenvalues γ_j^{-2} . Lemma 3.1 proves trace-class membership and the determinant identity. This gives statement 2, and differentiating the locally convergent product gives statement 3.

Conversely, if statement 3 holds, then for $|w| < \|K\|^{-1}$,

$$-\frac{D'_K(w)}{D_K(w)} = \mathrm{Tr}(K(\mathbf{I} - wK)^{-1}) = \sum_{n=0}^{\infty} \mathrm{Tr} K^{n+1} w^n = M(w). \quad (7.3)$$

Both determinants are normalized to one at the origin. Equality of logarithmic derivatives therefore gives $D_K = F$ near the origin and, by analyticity, on $\mathbb{C}$. Statement 2 implies RH because every zero of a positive Fredholm determinant lies on $(0, \infty)$. $\square$

The diagonal operator in the forward implication is a representation *conditional on RH*. It is not a source-derived construction and cannot be inserted as the first step in a proof. A genuine construction must specify K , including its measure, domain, boundary conditions, and normalization, from data independent of the assumed location of the arithmetic zeros. Theorem 7.1 then supplies an exact test of that construction.

7.2 Where the primes enter

Let $\Lambda(n)$ be the von Mangoldt function: $\Lambda(p^k) = \log p$ for primes p and integers $k \geq 1$, and $\Lambda(n) = 0$ otherwise. In the half-plane $\Re s > 1$, differentiating the absolutely convergent Euler product gives

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^s}. \quad (7.4)$$

Consequently, with $\psi_{\Gamma} = \Gamma'/\Gamma$,

$$\frac{\xi'(s)}{\xi(s)} = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi_{\Gamma}(s/2) - \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^s}. \quad (7.5)$$

Both identities are standard arithmetic consequences of Equation (3.1) [2]. The subscript on ψ_{Γ} prevents confusion with the Chebyshev prime-counting function.

Theorem 7.2 (Exact prime-trace closure). *RH is equivalent to the existence of a positive trace-class operator K satisfying, for every s in some nonempty open real interval $J \subset (1, \infty)$,*

$$2(s - \tfrac{1}{2}) \operatorname{Tr} \left[K(\mathbf{I} + (s - \tfrac{1}{2})^2 K)^{-1} \right] = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi_{\Gamma}(s/2) - \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^s}. \quad (7.6)$$

If the identity holds on such an interval, it holds for every real $s > 1$.

Proof. Assume the stated operator exists. Put $t = s - 1/2$ and define the entire function

$$R_K(s) = D_K(-(s - \tfrac{1}{2})^2).$$

For real $s > 1$, $R_K(s) > 0$, and Equation (7.3) gives

$$\frac{R'_K(s)}{R_K(s)} = 2t \operatorname{Tr} [K(\mathbf{I} + t^2 K)^{-1}].$$

Equations (7.5) and (7.6) imply $R_K(s) = c\xi(s)$ on J , where c is constant. The identity theorem extends this equality to all complex s . Evaluating at $s = 1/2$ gives $c = \xi(1/2)^{-1}$. Therefore $D_K = F$ on the negative real axis and hence everywhere. Theorem 7.1 proves RH. Conversely, the diagonal realization under RH satisfies Equation (7.3), and differentiating $D_K(-(s - 1/2)^2) = \xi(s)/\xi(1/2)$ yields Equation (7.6) for all real $s > 1$. $\square$

Equation (7.6) is the decisive arithmetic identification in this paper. Its left side is positive-operator geometry. Its right side is fixed by prime powers, the gamma factor, and the elementary completion factors. Neither side contains adjustable coefficients chosen from zero ordinates. The

identity makes precise what must be achieved beyond symmetry, leading asymptotics, or a generic self-adjoint construction.

The theorem also separates analytical continuation from divergent manipulation. The arithmetic side is evaluated only where the prime-power series converges absolutely. Equality of entire functions is obtained afterward by the identity theorem. Nothing requires multiplying the Euler factors on $\Re s = 1/2$, interchanging a divergent prime sum with a trace, or assuming the desired zero-free region in order to define the operator.

7.3 Why leading density cannot replace the identity

An eigenvalue counting law determines only an averaged spectral distribution. The full resolvent trace in Equation (7.6) determines much more: all inverse-power traces near the origin of the determinant, every zero with its multiplicity, and the normalization of the entire function. Two operators can have the same first several counting terms and still have different determinants. Section 10 supplies an explicit example at the precise zeta normalization.

The arithmetic interpretation is correspondingly exact. Under RH, the classical error estimate is $\pi(x) - \text{Li}(x) = O(\sqrt{x} \log x)$, with Li interpreted in its usual principal-value normalization [18]. The relevant fine-scale comparison is with $\text{Li}(x)$, not simply with $x/\log x$. The trace identity retains the complete prime-power information required by this connection, whereas matching the smooth counting term alone cannot recover it.

A continuous arithmetic intermediate is available in Appendix E. Proposition E.1 rewrites the same completed prime-side target as a Laplace transform, while Proposition E.4 connects it to a nonlocal characteristic ratio. These changes of representation retain the completion terms and do not replace the positive trace-class requirement in Theorem 7.2.

8 Transferring constitutive positivity to arithmetic

8.1 The exact bridge

The positive operator $O_\sigma = L_\sigma^* L_\sigma$ from Equation (2.8) suggests a precise sufficient condition. The condition is not an additional assertion that all arithmetic is physically generated by a Hilbert space. It specifies what an arithmetic representation of a particular source reduction would have to preserve.

Assumption 8.1 (Arithmetic Gram identification). For an admissible selector σ , there exists a complex-linear map

$$J_\sigma : \mathbb{C}[t] \longrightarrow \text{Dom}(L_\sigma) \quad (8.1)$$

such that, for all polynomials p, q ,

$$\langle L_\sigma J_\sigma p, L_\sigma J_\sigma q \rangle = \sum_{i,j} \overline{p_i} q_j m_{i+j}. \quad (8.2)$$

Here $p_i, q_j \in \mathbb{C}$ are polynomial coefficients, and the bar on p_i denotes complex conjugation. Both sides are conjugate-linear in p and linear in q ; in particular, replacing p by ip multiplies each side by $-i$. The source/readout norms and the map J_σ are fixed independently of any assumption that the zeta zeros lie on the critical line.

Theorem 8.2 (Constitutive Gram transfer). *A realization satisfying Assumption 8.1 proves RH. Alternatively, it is sufficient to construct a positive trace-class reduction K of the declared source/readout representation and prove Equation (7.2) in every degree.*

Proof. Setting $p = q$ in Equation (8.2) gives $q_\xi(p) = \|L_\sigma J_\sigma p\|^2 \geq 0$. Hence every H_N is positive semidefinite and Theorem 6.2 applies. The trace formulation follows from Theorem 7.1. $\square$

The proof is short because the difficult identification is displayed rather than hidden. The framework supplies a legitimate class of positive readout norms; it does not, merely by doing so, prove Equation (8.2). Constructing J_σ by first assuming that q_ξ is positive would reverse the intended implication. Likewise, choosing a Hilbert norm so that an already selected arithmetic matrix becomes a Gram matrix would encode the unproved target in the input [6].

For a positive trace-class K , the vectors $K^{n+1/2}$ are Hilbert–Schmidt operators and satisfy

$$\left\langle K^{i+1/2}, K^{j+1/2} \right\rangle_{\text{HS}} = \text{Tr } K^{i+j+1}. \quad (8.3)$$

Thus a successful trace construction produces the required polynomial Gram geometry explicitly. Equation (8.3) is useful in the forward direction, after K and its arithmetic trace have been derived. It is not a way to bypass the derivation of K .

8.2 Selector covariance

The framework organizes physically equivalent selectors by comparison arrows, not by the assertion that every boundary condition describes the same physical readout [4, 6]. Suppose an arrow $\kappa : \sigma \rightarrow \sigma'$ has unitary input and output maps U_κ, V_κ with the required domain transport, so that

$$L_{\sigma'} = V_\kappa L_\sigma U_\kappa^{-1}, \quad J_{\sigma'} = U_\kappa J_\sigma. \quad (8.4)$$

Proposition 8.3 (Naturality of arithmetic positivity). *Under Equation (8.4), an exact Gram identification for σ is an exact Gram identification for σ' .*

Proof. For any p, q , substitute Equation (8.4) into the left side of Equation (8.2). The input unitaries cancel, and the common output unitary preserves the inner product. The resulting expression equals its value for σ . $\square$

This is the appropriate invariant statement: the arithmetic form is preserved along the declared equivalence arrows. Changing to an inequivalent boundary phase or selecting a different potential is not automatically such an arrow. Requiring those distinct models to share one source rule is a transfer requirement; it is not a theorem that their determinants coincide. This distinction becomes concrete when the Dirichlet exponential model is tested below.

8.3 An infinite-dimensional requirement

Theorem 8.4 (Finite-rank obstruction). *No map satisfying Equation (8.2) can have finite-dimensional image under $L_\sigma J_\sigma$. In particular, one fixed finite-dimensional protected response sector cannot realize the full arithmetic moment tower.*

Proof. Suppose $\text{rank}(L_\sigma J_\sigma) = r < \infty$. For $N \geq r$, the $N + 1$ vectors $L_\sigma J_\sigma t^j$, $0 \leq j \leq N$, have a singular Gram matrix. Thus $\text{rank } H_N \leq r$. But Equation (8.2) implies RH by Theorem 8.2, and RH implies H_N is positive definite by Theorem 6.2. This contradicts $\text{rank } H_N < N + 1$. $\square$

The obstruction concerns representation rank, not source dimension. Function spaces over a four-dimensional manifold are normally infinite-dimensional. The theorem therefore excludes a finite arithmetic truncation masquerading as a complete representation, not a four-dimensional source. Finite sectors remain admissible for finite diagnostics, provided their rank limit and truncation error are reported rather than promoted into an all-degree proof.

8.4 The missing construction in operational form

An executable source route must determine three objects in the correct order: the source/readout Hilbert realization, an arithmetic insertion map such as J_σ , and an identity proving that its Gram coefficients equal the m_n extracted from ξ . A determinant route additionally needs a trace-class positive reduction K , with a declared relation to O_σ . There is no automatic rule $K = O_\sigma$, since an observability operator can be unbounded, have continuous spectrum, or fail every required trace condition [6].

The strongest compact statement is therefore

$$\boxed{\text{source-derived positive representation} + \text{exact arithmetic identification} \implies \text{RH.}} \quad (8.5)$$

The two premises are mathematically different. The remainder of the paper tests the extent to which a concrete geometric model can satisfy both.

8.5 A positive compatible-cochain sector

A concrete algebraic starting point is available without prescribing a one-dimensional potential. On the compatible cochain complex of Equation (2.12), fix an admissible reference phase and a positive weighted Hodge matrix $M_{2,\chi,r}$. With any Gate-T period sector held fixed, variation of the exact reciprocal component gives

$$C_r^{\text{rec}} = d_{1,r}^* M_{2,\chi,r} d_{1,r}, \quad a^* C_r^{\text{rec}} a = (d_{1,r} a)^* M_{2,\chi,r} (d_{1,r} a) \geq 0. \quad (8.6)$$

Here the star on a coefficient matrix denotes conjugate transpose; it is not an undeclared metric adjoint. If $M_{1,r} > 0$ is the one-cochain mass matrix, the Euclidean-normalized realization is $M_{1,r}^{-1/2} C_r^{\text{rec}} M_{1,r}^{-1/2}$. Its null space, period constraints, and boundary conditions must be treated explicitly. Only independently established redundancies may be removed [20, 6].

Equation (8.6) is a positive sector, not a positivity theorem for the full coupled foam Hessian. The phase potential, field couplings, singular boundary term, and admissible variation space must all be included before a total second variation is assessed. Nonnegative energy values do not imply a nonnegative Hessian at every reference configuration. Moreover, an energy Hessian is distinct from the physical-readout derivative construction in Equation (2.8) [21, 6].

8.6 Positive boundary response by interior elimination

Suppose the declared quadratic realization at refinement r , after its stated constraints, is a Hermitian matrix $C_r \geq 0$. Partition its variables into interior and retained boundary/interface coefficients:

$$C_r = \begin{pmatrix} C_{ii,r} & C_{ib,r} \\ C_{bi,r} & C_{bb,r} \end{pmatrix}, \quad C_{ii,r} > 0, \quad C_{bi,r} = C_{ib,r}^*. \quad (8.7)$$

The boundary contribution in Equation (2.13), when constructed, belongs in this same quadratic assembly. It is not added independently after arithmetic comparison. The following result is standard Schur-complement algebra; its role is to identify a positivity-preserving reduction compatible with the declared source construction, not to claim a new matrix theorem. Graph-theoretic Kron reduction supplies a familiar realization of this mechanism [23].

Proposition 8.5 (Positive interface reduction). *For Equation (8.7), define*

$$S_r = C_{bb,r} - C_{bi,r} C_{ii,r}^{-1} C_{ib,r}, \quad T_r b = \begin{pmatrix} -C_{ii,r}^{-1} C_{ib,r} b \\ b \end{pmatrix}. \quad (8.8)$$

Then $S_r = T_r^* C_r T_r \geq 0$, and

$$b^* S_r b = \min_a \begin{pmatrix} a \\ b \end{pmatrix}^* C_r \begin{pmatrix} a \\ b \end{pmatrix}. \quad (8.9)$$

If $C_r > 0$, then $S_r > 0$.

Proof. Complete the square:

$$\begin{pmatrix} a \\ b \end{pmatrix}^* C_r \begin{pmatrix} a \\ b \end{pmatrix} = \left(a + C_{ii,r}^{-1} C_{ib,r} b \right)^* C_{ii,r} \left(a + C_{ii,r}^{-1} C_{ib,r} b \right) + b^* S_r b.$$

The first term has its unique minimum zero at the stated interior extension. Substitution gives both the minimum identity and the congruence. If the full matrix is positive definite, the extension of any nonzero b is nonzero and has strictly positive quadratic value. $\square$

With a declared positive boundary mass matrix $M_{b,r}$, put

$$B_r = M_{b,r}^{-1/2} S_r M_{b,r}^{-1/2}, \quad K_r = B_r^{-1} \quad (8.10)$$

when $B_r > 0$ on the declared retained space. At each finite refinement this is a positive trace-class candidate. A continuum trace-class limit, dimensional normalization, and arithmetic identity remain independent requirements. A zero eigenvalue may not be removed, shifted, or assigned a pseudoinverse solely to force agreement with F ; any quotient, restriction, or shift changes the candidate and must be fixed by its source and readout definition.

The inverse interior block in Equation (8.8) can generate a dense boundary response from a sparse local assembly. This supplies a calculable route to nonlocality rather than an independently fitted nonlocal kernel. The boundary is an auxiliary interface of the mathematical representation. It is not automatically the physical Shadow or a complete holographic encoding. The separate selector and physical-registration maps remain required [20, 4].

Proposition 8.6 (Selector covariance of the reduced candidate). *Suppose equivalent selectors transport Equation (8.7) by $U_i \oplus U_b$, with U_i, U_b unitary, and transport the boundary mass by $M'_b = U_b M_b U_b^*$. Then*

$$S' = U_b S U_b^*, \quad B' = U_b B U_b^*, \quad K' = U_b K U_b^* \quad (8.11)$$

whenever the inverses exist. Their determinants and power traces agree.

Proof. Insert the blockwise conjugation into Equation (8.8); the interior unitaries cancel through C_{ii}^{-1} . Functional calculus transports the positive square root of M_b , and inversion gives the last relation. Trace and determinant are unitary invariants. $\square$

The block-compatible transport is a hypothesis to be checked, not a property of arbitrary selector changes. In the continuum, one may instead define a boundary form by minimizing a source form over extensions with prescribed trace. A densely defined closed nonnegative boundary form has an associated positive self-adjoint operator; a strictly positive lower bound and a trace-class inverse require further estimates [14]. Neither closability nor spectral compactness is supplied by formal elimination alone.

8.7 Static response is not the complete bulk determinant

The arithmetic object must be fixed before a spectral comparison. A static response operator and a parameter-dependent elimination of the complete source problem are different objects. Let $C_r > 0$ and $M_r > 0$, and define the full spectral pencil $P_r(w) = C_r - wM_r$. Wherever its interior block is invertible, write

$$S_r(w) = P_{bb,r}(w) - P_{bi,r}(w)P_{ii,r}(w)^{-1}P_{ib,r}(w). \quad (8.12)$$

Block elimination gives the exact factorization

$$\frac{\det(C_r - wM_r)}{\det C_r} = \frac{\det P_{ii,r}(w)}{\det C_{ii,r}} \frac{\det S_r(w)}{\det S_r(0)}. \quad (8.13)$$

To verify it, left-multiply $P_r(w)$ by the unit-determinant block triangular matrix that subtracts $P_{bi,r}(w)P_{ii,r}(w)^{-1}$ times its first block row from the second. The result is block triangular. Normalization uses the same calculation at $w = 0$.

The rational factors on the right of Equation (8.13) may have cancelling zeros and poles. The identity extends as the corresponding meromorphic identity; the product has the polynomial continuation on the left. Replacing the right side by $\det(I - wK_r)$ from the static boundary response generally discards both interior spectral data and the parameter dependence of $S_r(w)$. A complete bulk determinant must retain these factors. A static boundary determinant is a legitimate *different candidate*, but its equality to F must be proved in its own right. Neither determinant is the Riemann function merely because it descends from a positive source quadratic form.

8.8 Alternative arithmetic completion routes

Theorems 6.2 and 7.2 supply alternative conclusions of the construction. A positive trace-class realization of Equation (7.6) already proves the determinant identity; no separate construction of J_σ is then logically required for RH. Conversely, an all-degree positive representation of q_ξ proves RH without first realizing the proposed boundary operator. Section 12.1 develops a finite-to-infinite version of the latter route. These alternatives retain the same missing requirement: exact agreement with the arithmetic target, rather than positivity of an unrelated model.

9 Spectral growth required by the arithmetic target

9.1 Counting the squared ordinates

Suppose a positive determinant realization exists. Remove the kernel of K and set $A = K^{-1}$ on its natural dense domain. Its eigenvalues are γ_j^2 . The Riemann–von Mangoldt formula therefore

requires

$$\begin{aligned} N_A(E) &= \#\{j : \gamma_j^2 \leq E\} \\ &= \frac{\sqrt{E}}{4\pi} \log E - \frac{\log(2\pi) + 1}{2\pi} \sqrt{E} + O(\log E). \end{aligned} \quad (9.1)$$

This follows by substituting $T = \sqrt{E}$ into $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$ [2]. Multiplicity is retained and only positive ordinates are counted. Doubling to include both signs would change every leading coefficient.

Proposition 9.1 (Compact-elliptic obstruction). *The full spectrum of a positive classical elliptic operator of fixed order $m > 0$ on a smooth compact d -dimensional manifold, under the usual regular elliptic boundary conditions when a boundary is present, cannot equal the squared zeta ordinates in a positive determinant realization. The same conclusion holds after a fixed positive affine spectral rescaling or a perturbation affecting only finitely many eigenvalues.*

Proof. For this operator class the leading Weyl law is $N(E) \sim cE^{d/m}$, with $c > 0$ determined by the principal symbol [15]. Equation (9.1) instead has leading term $\frac{1}{4\pi}E^{1/2} \log E$. If $d/m \neq 1/2$, the ratio of the two expressions tends to zero or infinity. If $d/m = 1/2$, the logarithm still makes the ratio unbounded. An affine spectral rescaling changes coefficients but not this distinction, and finitely many eigenvalue changes alter $N(E)$ by $O(1)$. $\square$

For a four-dimensional regular source chart, choosing order eight would match the power $E^{1/2}$ but not the logarithm. For a three-dimensional regular Shadow chart, order six has the same limitation. This is a restriction on an entire-spectrum realization in a specified operator class, not a dimensional refutation of Counterspace. Singular strata, noncompact corridors, nonlocal operators, logarithmic symbols, or independently selected infinite-dimensional subspaces fall outside the stated proposition. Each alternative requires its own counting theorem; the existence of singular source structure alone does not supply the required coefficient [5].

9.2 The heat-trace fingerprint

The counting law determines an additional useful target. Here $\tau > 0$ is an analytical heat parameter, not the operational clock of Equation (2.2).

Proposition 9.2 (Required heat asymptotics). *Any positive determinant realization has*

$$\text{Tr } e^{-\tau A} = \frac{\log(1/\tau) - \gamma_E - 2\log(4\pi)}{8\sqrt{\pi\tau}} + O(\log(1/\tau)) \quad (\tau \downarrow 0), \quad (9.2)$$

where γ_E is Euler's constant.

Proof. Integration by parts gives

$$\text{Tr } e^{-\tau A} = \tau \int_0^\infty e^{-\tau E} N_A(E) dE.$$

Insert Equation (9.1). With $v = \tau E$, the first two terms use $\Gamma(3/2) = \sqrt{\pi}/2$ and $\Gamma'(3/2)/\Gamma(3/2) = 2 - \gamma_E - 2\log 2$. Their coefficients simplify to Equation (9.2). Integrating the $O(\log E)$ remainder gives $O(\log(1/\tau))$; the bounded low-energy interval contributes only a bounded error. $\square$

The leading logarithmic heat term is a stringent geometric requirement. It distinguishes the arithmetic target from an ordinary compact-manifold heat expansion even before individual eigenvalues are compared. It remains only a necessary condition: the exponential model below has the same first two counting terms and therefore the same displayed heat terms without having the correct determinant.

There is also a trace-ideal consequence. Inverting the leading zero-counting relation gives $\gamma_j \sim 2\pi j / \log j$. Thus

$$\lambda_j(K) \sim \left(\frac{\log j}{2\pi j} \right)^2, \quad \text{Tr } K^q < \infty \iff q > \frac{1}{2}. \quad (9.3)$$

The borderline series at $q = 1/2$ behaves like $\sum (\log j)/j$ and diverges. This calculation specifies how much compactness the arithmetic realization needs, rather than treating trace-class membership as an unspecified technical assumption.

9.3 A smooth boundary response does not evade the obstruction

Nonlocality by itself does not imply an anomalous leading counting law. For a smooth compact boundary, the usual Dirichlet-to-Neumann operator is a classical elliptic pseudodifferential operator of order one, and its spectrum is the Steklov spectrum [24]. Its boundary dimension, operator order, and mass measure therefore remain decisive.

Proposition 9.3 (Regular boundary-power obstruction). *Let B be such a positive boundary operator on a compact three-dimensional boundary, restricted off a finite-dimensional kernel when necessary, with eigenvalues $b_j \sim cj^{1/3}$, $c > 0$. For any fixed $\alpha > 0$, $B^{-\alpha}$ is trace class exactly when $\alpha > 3$. No positive affine rescaling of B^α has the full counting law in Equation (9.1).*

Proof. The inverse-power series is comparable to $\sum_j j^{-\alpha/3}$, giving the trace-class condition. The counting function of B^α is asymptotic to a positive constant times $E^{3/\alpha}$. At $\alpha = 6$ its power matches $E^{1/2}$ but the logarithm is absent; other powers fail already at the exponent. A positive affine rescaling does not change either conclusion. $\square$

Thus an ordinary smooth boundary operator with a convenient inverse power cannot be promoted into the arithmetic realization. Singular interfaces, noncompact limits, or nonclassical spectral rules are possibilities to be analyzed, not guaranteed solutions. The compact-elliptic and calibrated Morse exclusions remain restricted to their stated classes; they do not exclude every smooth potential or all conventional spectral mathematics. In particular, the noncompact smooth exponential model below already reproduces the logarithmic density while failing the exact determinant.

10 An explicit exponential-corridor construction

10.1 The positive operator

Consider the differential expression

$$A_{\text{exp}} = -\frac{d^2}{du^2} + 16\pi^2 e^{4u} \quad \text{on } L^2([0, \infty), du), \quad f(0) = 0. \quad (10.1)$$

Its coefficient is calibrated to the required asymptotic density, not derived from the ECL. The variable u is an auxiliary logarithmic corridor coordinate. No physical one-dimensional source or

ontic time is asserted. This example tests whether a particularly economical noncompact realization can satisfy the arithmetic conditions left open by the compact-elliptic obstruction.

Define the closed quadratic form

$$a_{\text{exp}}[f] = \int_0^\infty \left(|f'(u)|^2 + 16\pi^2 e^{4u} |f(u)|^2 \right) du, \quad (10.2)$$

with form domain $H_0^1([0, \infty)) \cap L^2([0, \infty), e^{4u} du)$. Its associated Friedrichs operator is the realization intended in Equation (10.1).

Proposition 10.1 (Well-defined positive corridor). *A_{exp} is self-adjoint, satisfies $A_{\text{exp}} \geq 16\pi^2 \mathbf{I}$, and has compact resolvent. Its eigenvalues are positive, discrete, and simple.*

Proof. The densely defined closed form is bounded below by $16\pi^2 \|f\|_2^2$, so the representation theorem for semibounded forms gives a self-adjoint operator with that bound [14]. To prove compactness, consider a set bounded in the form norm. On each interval $[0, R]$, the H^1 bound gives compactness in L^2 . The tail satisfies

$$\int_R^\infty |f(u)|^2 du \leq \frac{e^{-4R}}{16\pi^2} a_{\text{exp}}[f],$$

so it is uniformly small. A diagonal subsequence argument gives compactness of the full form-domain embedding, hence compact resolvent. Simplicity follows from the Wronskian: two square-integrable solutions at infinity with the same eigenvalue are proportional, and the Dirichlet condition supplies only one independent endpoint condition. $\square$

10.2 An exactly computable counting law

For $E > 16\pi^2$, let $\delta = 4\pi/\sqrt{E}$ and let $u_E = \frac{1}{4} \log(E/(16\pi^2))$ be the turning point. The classical one-dimensional phase integral is

$$\begin{aligned} I_{\text{exp}}(E) &= \frac{1}{\pi} \int_0^{u_E} \sqrt{E - 16\pi^2 e^{4u}} du \\ &= \frac{\sqrt{E}}{2\pi} \left[\log \frac{1 + \sqrt{1 - \delta^2}}{\delta} - \sqrt{1 - \delta^2} \right]. \end{aligned} \quad (10.3)$$

The second equality follows from the substitution $v = 4\pi e^{2u}/\sqrt{E}$ and integration of $\sqrt{1 - v^2}/v$ between δ and one.

For this smooth monotone exponential potential, the one-dimensional counting theorem gives $N_{\text{exp}}(E) = I_{\text{exp}}(E) + O(1)$. It is also a direct rescaling of the Morse half-line counting result: setting $r = 2u + \log(4\pi)$ gives four times the operator $-d^2/dr^2 + \frac{1}{4}e^{2r}$ on $r \geq \log(4\pi)$ [16]. Expanding Equation (10.3) therefore gives

$$N_{\text{exp}}(E) = \frac{\sqrt{E}}{2\pi} \left(\log \frac{\sqrt{E}}{2\pi} - 1 \right) + O(1). \quad (10.4)$$

Both displayed coefficients agree with Equation (9.1). The available classically allowed corridor length grows logarithmically with E , which is the explicit mechanism generating the extra logarithm absent from the compact Weyl law.

This count implies $\sum_j E_j^{-1} < \infty$, for example by integrating $E^{-1} dN_{\text{exp}}(E)$ at infinity. Thus A_{exp}^{-1} is positive trace class. The construction has now met positivity, discreteness, trace-class compactness, and the first two arithmetic counting terms. It remains to test the entire determinant.

10.3 The Bessel characteristic determinant

Put $y = 2\pi e^{2u}$. The eigenvalue equation reduces to

$$y^2 f_{yy} + y f_y + \left(\frac{E}{4} - y^2 \right) f = 0. \quad (10.5)$$

The solution decaying at infinity is proportional to the modified Bessel function $K_{i\sqrt{E}/2}(y)$. Consequently the Dirichlet characteristic function is

$$D_{\text{exp}}(w) = \frac{K_{i\sqrt{w}/2}(2\pi)}{K_0(2\pi)}. \quad (10.6)$$

The evenness of K_ν in its order makes this an entire function of w . Bessel-order zeros and their spectral interpretation have classical antecedents; this is a concrete normalization of that family [16, 17].

Proposition 10.2 (Exact determinant of the corridor). *The characteristic function satisfies*

$$D_{\text{exp}}(w) = \det(\mathbf{I} - w A_{\text{exp}}^{-1}). \quad (10.7)$$

Proof. A zero of the numerator in Equation (10.6) supplies a decaying Dirichlet solution and is therefore an eigenvalue. Conversely each eigenvalue gives such a zero. The zeros are simple: differentiating the differential equation in w and applying Green's identity at an eigenvalue E gives

$$\int_0^\infty f(u, E)^2 du = -f_u(0, E) \partial_w f(0, E).$$

The left side is positive, and $f_u(0, E) \neq 0$ by uniqueness for the second-order equation. Thus the boundary characteristic has nonzero derivative.

The integral $K_{i\sqrt{w}/2}(2\pi) = \int_0^\infty e^{-2\pi \cosh v} \cos(\sqrt{w} v/2) dv$ bounds its maximum modulus on $|w| \leq R$ by $K_{\sqrt{R}/2}(2\pi)$. The large-order bound is $\log K_{\sqrt{R}/2}(2\pi) = O(\sqrt{R} \log R)$, so the entire order is at most $1/2$ [17]. The counting law forces order $1/2$. Its normalized genus-zero product is therefore exactly the product over eigenvalues in Equation (10.7). $\square$

10.4 The determinant fails despite the correct density

Theorem 10.3 (Exponential determinant mismatch). *For real $s \rightarrow +\infty$,*

$$\frac{F(-(s - \frac{1}{2})^2)}{D_{\text{exp}}(-(s - \frac{1}{2})^2)} \sim \frac{K_0(2\pi)}{\xi(\frac{1}{2})(2\pi)^{1/4}} s^{9/4}. \quad (10.8)$$

In particular $D_{\text{exp}} \neq F$, and A_{exp}^{-1} does not satisfy the prime-trace identity.

Proof. Write $\nu = s/2 - 1/4$. At fixed positive argument, the Bessel integral can be written

$$K_\nu(2\pi) = \frac{1}{2} \pi^{-\nu} \int_0^\infty y^{\nu-1} e^{-y-\pi^2/y} dy.$$

For a gamma random variable Y of shape $\nu > 1$ and unit scale, this gives

$$\frac{K_\nu(2\pi)}{\frac{1}{2} \pi^{-\nu} \Gamma(\nu)} = \mathbb{E} e^{-\pi^2/Y}.$$

Since $0 \leq 1 - e^{-v} \leq v$,

$$1 - \frac{\pi^2}{\nu - 1} \leq \mathbb{E}e^{-\pi^2/Y} \leq 1. \quad (10.9)$$

Thus $K_\nu(2\pi) \sim \frac{1}{2}\pi^{-\nu}\Gamma(\nu)$; the same standard asymptotic is recorded in the Bessel literature [17].

For real $s \rightarrow \infty$, $\zeta(s) = 1 + O(2^{-s})$, so Equation (3.2) gives

$$\xi(s) \sim \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

Use $F(-(s-1/2)^2) = \xi(s)/\xi(1/2)$, the evenness of Bessel order, and $\Gamma(s/2)/\Gamma(s/2-1/4) \sim (s/2)^{1/4}$. The quotient simplifies to Equation (10.8). An identity of determinants would make the quotient identically one, which is impossible. The final assertion follows from Theorem 7.2. $\square$

This failure is stronger than a numerical disagreement in a low eigenvalue. It is an analytical obstruction in an asymptotic regime where all quantities are real and positive. The model does not merely need a better estimate of its existing spectrum: its characteristic function has the wrong normalization-dependent power. The next construction tests whether a minimal additional potential term can repair this defect.

11 A Morse correction and a family-level obstruction

11.1 The one-parameter repair

Consider the calibrated Dirichlet family

$$A_\kappa = -\frac{d^2}{du^2} + 16\pi^2 e^{4u} - 16\pi\kappa e^{2u}, \quad u \geq 0, \quad f(0) = 0, \quad \kappa < \pi. \quad (11.1)$$

Here κ is a model parameter, not a Counterspace invariant. For $y = e^{2u} \geq 1$, the potential is $16\pi y(\pi y - \kappa)$; it is increasing and bounded below by $16\pi(\pi - \kappa) > 0$. The form argument used for Proposition 10.1 therefore gives a positive self-adjoint operator with compact resolvent.

Under $r = 2u + \log(4\pi)$, Equation (11.1) becomes four times the standard Morse operator with potential $\frac{1}{4}e^{2r} - \kappa e^r$.¹ Its eigenvalue count retains Equation (10.4), with a κ -dependent bounded remainder, and its inverse is trace class [16]. The decaying solution is proportional to

$$(4\pi e^{2u})^{-1/2} W_{\kappa, i\sqrt{w}/2}(4\pi e^{2u}), \quad (11.2)$$

where W is the Whittaker function. The normalized determinant is

$$D_\kappa(w) = \frac{W_{\kappa, i\sqrt{w}/2}(4\pi)}{W_{\kappa, 0}(4\pi)} = \det(I - wA_\kappa^{-1}). \quad (11.3)$$

The denominator is nonzero because zero is not an eigenvalue. Lemma 11.4 below supplies an explicit bound throughout the complex order plane, and Proposition 11.5 proves the simple-zero and normalized genus-zero product statements, for this Dirichlet case and for fixed Robin data. The real-order asymptotic of Lemma 11.1 is used only for the subsequent comparison, not as a complex-plane growth bound.

¹Lagarias writes $V_k(r) = \frac{1}{4}e^{2r} + ke^r$, so his parameter is $k = -\kappa$ in this normalized expression, and the decaying solution uses $W_{-k, \nu} = W_{\kappa, \nu}$. The spectral factor four and the change of endpoint must also be retained when applying his counting theorem [16, Theorems 2.1 and 3.2].

11.2 The necessary asymptotic correction

The following coefficient, derived in Appendix A, is sufficient to test every member of the family.

Lemma 11.1 (Fixed-argument Whittaker expansion). *For fixed $x > 0$ and fixed real κ , as $\mu \rightarrow +\infty$,*

$$W_{\kappa,\mu}(x) = \frac{\sqrt{x}}{2\sqrt{\pi}} \Gamma(\mu)(x/4)^{-\mu} \mu^\kappa \left[1 + \frac{c(\kappa, x)}{\mu} + O(\mu^{-2}) \right], \quad (11.4)$$

$$c(\kappa, x) = -\frac{\kappa^2}{2} + \frac{\kappa x}{2} - \frac{x^2}{16}. \quad (11.5)$$

Theorem 11.2 (Calibrated Dirichlet Morse-family exclusion). *No member of Equation (11.1), for $\kappa < \pi$, has $D_\kappa = F$. More precisely, with $\mu = s/2 - 1/4$,*

$$\frac{F(-(s - \frac{1}{2})^2)}{D_\kappa(-(s - \frac{1}{2})^2)} = C_\kappa \mu^{9/4-\kappa} \left[1 + \frac{\delta_\kappa}{\mu} + O(\mu^{-2}) \right], \quad (11.6)$$

where

$$C_\kappa = \frac{2W_{\kappa,0}(4\pi)}{\xi(\frac{1}{2})\pi^{1/4}} > 0, \quad \delta_\kappa = \frac{\kappa^2}{2} - 2\pi\kappa + \pi^2 - \frac{3}{32}. \quad (11.7)$$

The power mismatch can vanish only at $\kappa = 9/4$, but then

$$\delta_{9/4} = \pi^2 - \frac{9}{2}\pi + \frac{39}{16} < 0, \quad (11.8)$$

which prevents the quotient from being identically one.

Proof. At $s = 2\mu + 1/2$, the defining completion gives

$$\xi(s) = \left(2\mu^2 - \frac{1}{8} \right) \pi^{-\mu-1/4} \Gamma(\mu + 1/4) \zeta(2\mu + 1/2).$$

The gamma-ratio expansion and the exponentially small zeta correction yield

$$\xi(2\mu + \frac{1}{2}) = 2\pi^{-1/4} \Gamma(\mu) \pi^{-\mu} \mu^{9/4} \left[1 - \frac{3}{32\mu} + O(\mu^{-2}) \right]. \quad (11.9)$$

At $x = 4\pi$, Lemma 11.1 has leading prefactor $\Gamma(\mu)\pi^{-\mu}\mu^\kappa$ and correction $-\kappa^2/2 + 2\pi\kappa - \pi^2$. Dividing Equation (11.9) by that expansion, with the normalizations in Equation (11.3), proves Equations (11.6)–(11.7).

For $\kappa \neq 9/4$, the quotient tends to zero or infinity. For $\kappa = 9/4$, an identical quotient would first require $C_{9/4} = 1$ and then $\delta_{9/4} = 0$. Equation (11.8) rules out the second requirement, independently of the value of the first. Its sign follows, for example, from $3 < \pi < 22/7$. Finally, $W_{\kappa,0}(4\pi) > 0$: its decaying real solution is positive at infinity and cannot have a zero on the half-line, since such a zero would produce a zero-energy Dirichlet eigenfunction on a shorter interval with strictly positive potential. Hence $C_\kappa > 0$. $\square$

The family-level conclusion is deliberately bounded. It excludes the calibrated exponential-plus-one-Morse-term potential with the stated Dirichlet endpoint, not every singular Sturm–Liouville system, every boundary condition, or every source-derived operator. Its strength is that it tests a natural attempted repair rather than stopping at the first mismatch. The correction $\kappa = 9/4$

removes the leading power defect but cannot reproduce the next coefficient of the actual arithmetic completion.

At the framework level, this provides a precise example of the distinction between a transferable constitutive map and an expressive fitting family. The parameter was introduced to repair a known target asymptotic. Even a successful repair would not have been a source derivation without an independent rule selecting it. The stronger result obtained here is that this repair fails the arithmetic identity on its own mathematical terms. Neither source completeness nor the positivity of the potential changes the nonzero coefficient in Equation (11.8).

11.3 Free potential scale and endpoint

The Dirichlet endpoint and exponential scale need not be fixed independently. With the kinetic coefficient and exponential rate retained as in Equation (11.1), consider

$$A_{c,\kappa,u_0} = -\frac{d^2}{du^2} + c^2 e^{4u} - 4c\kappa e^{2u}, \quad u \in [u_0, \infty), \quad f(u_0) = 0, \quad c > 0, \quad \kappa, u_0 \in \mathbb{R}. \quad (11.10)$$

The change of coordinate $r = 2u + \log c$, including the unitary factor $2^{-1/2}$ on the wave function, gives

$$A_{c,\kappa,u_0} \simeq 4 \left(-\frac{d^2}{dr^2} + \frac{1}{4} e^{2r} - \kappa e^r \right), \quad r \geq \log x, \quad x := ce^{2u_0} > 0. \quad (11.11)$$

Thus scale and endpoint are not two independent spectral adjustments: for fixed κ , their only spectral combination is x . The classical Morse counting law [16, Theorem 3.2], with the factor four in Equation (11.11), reads

$$N_{c,\kappa,u_0}(E) = \frac{\sqrt{E}}{2\pi} \left[\log \sqrt{E} + \log \frac{2}{x} - 1 \right] + O(1). \quad (11.12)$$

Comparison with Equation (9.1) forces

$$\log \frac{2}{x} = -\log(2\pi), \quad x = ce^{2u_0} = 4\pi. \quad (11.13)$$

The $O(1)$ correction depends on the fixed parameters but cannot change the coefficient of $\sqrt{E}$.

Corollary 11.3 (Scale–endpoint closure of the Dirichlet Morse exclusion). *No strictly positive Dirichlet realization of Equation (11.10) has normalized determinant F . Hence freeing the potential scale and endpoint does not repair the exclusion in Theorem 11.2.*

Proof. Every operator in Equation (11.10) is semibounded with compact resolvent, since its potential is $(ce^{2u} - 2\kappa)^2 - 4\kappa^2$ and tends to positive infinity. In the strictly positive case, Equation (11.12) implies that the inverse is trace class. The same characteristic-function argument as in Equation (11.3) gives

$$D_{\kappa,x}(w) = \frac{W_{\kappa,i\sqrt{w}/2}(x)}{W_{\kappa,0}(x)} = \det(I - wA_{c,\kappa,u_0}^{-1}). \quad (11.14)$$

The denominator is nonzero. Put $\mu = s/2 - 1/4$ for real $s \rightarrow +\infty$. Lemma 11.1 and Equation (11.9) yield

$$\frac{F(-(s - \frac{1}{2})^2)}{D_{\kappa,x}(-(s - \frac{1}{2})^2)} = C_{\kappa,x} \left(\frac{x}{4\pi} \right)^\mu \mu^{9/4-\kappa} \left[1 + \frac{\delta_\kappa(x)}{\mu} + O(\mu^{-2}) \right], \quad (11.15)$$

$$C_{\kappa,x} = \frac{4\pi^{1/4} W_{\kappa,0}(x)}{\sqrt{x} \xi(\frac{1}{2})}, \quad \delta_\kappa(x) = \frac{\kappa^2}{2} - \frac{\kappa x}{2} + \frac{x^2}{16} - \frac{3}{32}. \quad (11.16)$$

The constant is positive in the strictly positive realization, as also follows by comparing the positive determinants on the negative real axis. An identity of determinants first forces $x = 4\pi$ by the exponential factor, consistently with the independent counting argument. It then forces $\kappa = 9/4$ by the power factor. At that value,

$$\delta_{9/4}(x) = \frac{x^2 - 18x + 39}{16}, \quad \delta_{9/4}(x) = 0 \iff x = 9 \pm \sqrt{42}. \quad (11.17)$$

Neither root equals 4π , where Equation (11.8) gives the nonzero coefficient already computed. Thus equality is impossible, regardless of the leading normalization constant. $\square$

The two roots are approximately 2.5192593 and 15.4807407. Canceling the inverse-order coefficient at either root breaks the necessary second counting coefficient and leaves the exponential factor in Equation (11.15). At fixed $x = 4\pi$, changing c and compensating with u_0 merely gives a unitarily equivalent operator. The exclusion concerns this full scale–endpoint Dirichlet family with the stated spectral normalization, not arbitrary boundary conditions or additional potentials. A realization with a zero or negative eigenvalue cannot supply a positive inverse determinant with zero set $(0, \infty)$; deleting such modes or adding a spectral shift defines a different candidate and is not covered by Corollary 11.3 alone. Constant spectral shifts and all fixed real separated endpoint conditions are addressed explicitly in Sections 11.5 and 11.6; deletion of modes remains outside those statements.

11.4 Complex-order growth and canonical determinant normalization

A real-order asymptotic does not control an entire function in the complex spectral plane. The determinant identification in Equations (11.3) and (11.14) requires the latter control. The following estimate displays it and also covers a fixed Robin endpoint. The special-function inputs are the Tricomi integral, the Whittaker–Tricomi relation, evenness in the order, and the reciprocal-gamma product [17, Sections 5.8, 13.4, 13.14, and 13.16]; no zero-location assumption about ξ is used.

Lemma 11.4 (Uniform complex-order growth). *Fix $\kappa \in \mathbb{R}$ and $0 < x_- < x_+ < \infty$. There is a constant C such that, for $R \geq 1$,*

$$\sup_{\substack{|\nu| \leq R \\ x \in [x_-, x_+]}} (|W_{\kappa, \nu}(x)| + |\partial_x W_{\kappa, \nu}(x)|) \leq \exp(C(1 + R) \log(2 + R)). \quad (11.18)$$

Consequently $W_{\kappa, i\sqrt{w}/2}(x)$ and $2x\partial_x W_{\kappa, i\sqrt{w}/2}(x) - (1 + h)W_{\kappa, i\sqrt{w}/2}(x)$, for fixed real h , are entire functions of w of order at most $1/2$. The square-root notation denotes their even power series.

Proof. Choose an integer $m \geq 0$ such that $b_0 = m + 1/2 - \kappa > 0$, and put $a = \nu + 1/2 - \kappa$ and $b = \nu + \kappa - 1/2$. For $\Re a > 0$, the standard integral and m integrations by parts give

$$\begin{aligned} W_{\kappa, \nu}(x) &= \frac{e^{-x/2} x^{\nu+1/2}}{\Gamma(a)} \int_0^\infty t^{a-1} e^{-xt} (1+t)^b dt \\ &= \frac{(-1)^m e^{-x/2} x^{\nu+1/2}}{\Gamma(a+m)} \int_0^\infty t^{a+m-1} \frac{d^m}{dt^m} [e^{-xt} (1+t)^b] dt. \end{aligned} \quad (11.19)$$

The boundary terms vanish in the initial half-plane. The last integral is holomorphic whenever $\Re \nu > -b_0$, so analytic continuation makes it valid throughout $\Re \nu \geq 0$, including cases where the

first integral is not convergent. Its differentiated factor is the finite sum

$$\frac{d^m}{dt^m} [e^{-xt}(1+t)^b] = e^{-xt} \sum_{j=0}^m \binom{m}{j} (-x)^{m-j} b^j (1+t)^{b-j}, \quad b^j = \prod_{\ell=0}^{j-1} (b-\ell). \quad (11.20)$$

For $p = \Re \nu \in [0, R]$, its coefficients have magnitude at most $C_1(1+R)^m$. Splitting the integral at $t = 1$ bounds the remaining absolute integral, uniformly in x , by

$$C_2 2^R \left(1 + x_-^{-2R-B} \Gamma(2R+B)\right) \quad (11.21)$$

for a fixed sufficiently large $B > 0$. On $(0, 1)$ one uses $t^{p+b_0-1} \leq t^{b_0-1}$; on $[1, \infty)$ the powers are bounded by a fixed multiple of $2^R t^{2R+B-1}$. The logarithm of the right side is $O((1+R) \log(2+R))$.

The remaining reciprocal gamma factor has the same required bound. Indeed,

$$\Gamma(z)^{-1} = ze^{\gamma_E z} \prod_{n=1}^{\infty} (1 + z/n) e^{-z/n}. \quad (11.22)$$

For $|z| \leq R + C_3$, split the product at an integer exceeding $2(R + C_3)$. The logarithm of the modulus of the finite part is bounded above by $\sum [\log(1 + |z|/n) + |z|/n] = O((1+R) \log(2+R))$. The tail is bounded by $C_4 |z|^2 \sum n^{-2} = O(1+R)$, using $|z/n| \leq 1/2$. Zeros of the reciprocal gamma function cause no difficulty for this upper bound. Also $|x^{\nu+1/2}| \leq C_5 \exp(C_6 R)$.

Differentiating Equation (11.19) in x only adds a factor bounded by $C_7(1+R)$ outside the integral and finitely many terms with an extra factor t inside it. These are bounded by increasing B in Equation (11.21). Differentiation under the integral is justified by the same exponential majorant on the compact x -interval. This proves Equation (11.18) in the right half-plane. The standard evenness $W_{\kappa, -\nu} = W_{\kappa, \nu}$, also after x -differentiation, supplies the other half-plane. Entire dependence on ν and this evenness give entire dependence on w ; substituting $|\nu| \leq \sqrt{|w|}/2$ gives order at most $1/2$. $\square$

Proposition 11.5 (Dirichlet and Robin characteristic products, including shifts). *Let A be the realization of*

$$-\frac{d^2}{du^2} + c^2 e^{4u} - 4c\kappa e^{2u} + \lambda, \quad u \geq u_0, \quad (11.23)$$

with $c > 0$, real fixed κ, u_0, λ , and either $f(u_0) = 0$ or $f'(u_0) = hf(u_0)$ for a fixed $h \in \mathbb{R}$. Assume A is strictly positive, and set $x = ce^{2u_0}$. Define the unshifted characteristic functions

$$\begin{aligned} \mathcal{H}_D(w) &= W_{\kappa, i\sqrt{w}/2}(x), \\ \mathcal{H}_R(w) &= Q_{\kappa, h, i\sqrt{w}/2}(x), \quad Q_{\kappa, h, \nu}(x) = 2x \partial_x W_{\kappa, \nu}(x) - (1+h)W_{\kappa, \nu}(x). \end{aligned} \quad (11.24)$$

For the chosen boundary condition $B \in \{D, R\}$, one has

$$\frac{\mathcal{H}_B(w - \lambda)}{\mathcal{H}_B(-\lambda)} = \prod_{j=1}^{\infty} (1 - w/E_j) = \det(I - wA^{-1}), \quad (11.25)$$

where $E_j > 0$ are the eigenvalues of A , without deletion. The product has genus zero and the entire function has order $1/2$.

Proof. The real separated half-line Morse problem is self-adjoint, has compact resolvent, and has simple discrete spectrum. For every fixed endpoint condition its counting law is Equation (11.12) [16, Theorems 2.1 and 3.2]. A constant spectral shift leaves its two displayed coefficients unchanged, so $\sum E_j^{-1} < \infty$ under the stated positivity assumption.

With $y = ce^{2u}$, the decaying solution is $f(u, w) = y^{-1/2}W_{\kappa, \nu}(y)$ with $\nu^2 = (\lambda - w)/4$. Its Dirichlet boundary value is $x^{-1/2}\mathcal{H}_D(w - \lambda)$; its Robin residual is $x^{-1/2}\mathcal{H}_R(w - \lambda)$. Thus each characteristic zero is an eigenvalue and conversely. To check analytic multiplicity, take the real decaying solution at a real eigenvalue E and put $g = \partial_w f$. Let $\mathcal{L}$ denote the differential expression in Equation (11.23). Differentiating before imposing any endpoint condition on g gives $(\mathcal{L} - E)g = f$. Green's identity gives

$$\int_{u_0}^{\infty} f(u, E)^2 du = f(u_0, E)g'(u_0, E) - f'(u_0, E)g(u_0, E). \quad (11.26)$$

The terms at infinity vanish by the decaying Whittaker asymptotic and its parameter derivative. For Dirichlet data this equals $-f'(u_0, E)g(u_0, E)$; for Robin data it equals $f(u_0, E)(g'(u_0, E) - hg(u_0, E))$. The nonzero endpoint factor in either case cannot vanish by uniqueness for the differential equation. The positive integral therefore proves that the characteristic zero is simple.

Lemma 11.4 bounds the full complex-plane growth by $\exp(O(\sqrt{|w|}\log(2 + |w|)))$; a fixed translation $w \mapsto w - \lambda$ preserves this order bound. Hadamard factorization for order less than one supplies no nonconstant exponential factor. Since zero is not an eigenvalue, division by $\mathcal{H}_B(-\lambda)$ fixes the remaining constant to one. The trace-class product is exactly Equation (11.25). The eigenvalue count gives the matching lower bound on entire order. $\square$

11.5 Constant spectral shifts

A fixed spectral shift is invisible to the first two counting coefficients but not to the full determinant. More precisely, the change of the *smooth counting main term* under $E \mapsto E - \lambda$ is $O(E^{-1/2} \log E)$. This is not a vanishing bound on the difference of two integer-valued counting staircases; the unchanged $O(1)$ counting remainder must still be retained.

Corollary 11.6 (Constant-shift Dirichlet exclusion). *No strictly positive Dirichlet realization of Equation (11.23), for any real constant λ , has normalized inverse determinant F .*

Proof. Write $s = 2\mu + 1/2$. At $w = -4\mu^2$, the shifted Whittaker order is

$$\mu_\lambda = \sqrt{\mu^2 + \lambda/4} = \mu + \frac{\lambda}{8\mu} + O(\mu^{-3}). \quad (11.27)$$

The gamma-ratio expansion, or the mean-value formula with $\psi_\Gamma(\mu) = \log \mu + O(\mu^{-1})$, gives

$$\log \frac{\Gamma(\mu_\lambda)(x/4)^{-\mu_\lambda} \mu_\lambda^\kappa}{\Gamma(\mu)(x/4)^{-\mu} \mu^\kappa} = \frac{\lambda}{8\mu} (\log \mu - \log(x/4)) + O(\mu^{-2}). \quad (11.28)$$

Apply Lemma 11.1 and Proposition 11.5. Equality to F first forces $x = 4\pi$ and $\kappa = 9/4$ by the exponential and power factors, respectively, and fixes the leading normalization. The remaining logarithm of the quotient is

$$-\frac{\lambda}{8\mu} (\log \mu - \log \pi) + \frac{\delta_{9/4}(4\pi)}{\mu} + O(\mu^{-2}). \quad (11.29)$$

Multiplication by $\mu/\log \mu$ forces $\lambda = 0$. The nonzero coefficient in Equation (11.8) then contradicts equality. The initial normalization is $\mathcal{H}_D(-\lambda)$, not $\mathcal{H}_D(0)$, so the argument also covers shifts of unshifted operators having a zero or negative eigenvalue. $\square$

11.6 Robin matching and an analytic second-order obstruction

Robin data are not covered by the Dirichlet power comparison. The derivative in Equation (11.24) supplies an additional power of the large order. To track it beyond leading order, define

$$L_{\kappa,\mu}(x) = \frac{\sqrt{x}}{2\sqrt{\pi}} \Gamma(\mu) (x/4)^{-\mu} \mu^\kappa. \quad (11.30)$$

Appendix A.1 proves the locally x -uniform expansion, including its first x -derivative,

$$\begin{aligned} \log \frac{W_{\kappa,\mu}(x)}{L_{\kappa,\mu}(x)} &= \frac{c(\kappa, x)}{\mu} + \frac{c_2(\kappa, x)}{\mu^2} + O(\mu^{-3}), \\ c_2(\kappa, x) &= -\frac{\kappa^3}{6} + \frac{\kappa}{24} + \frac{\kappa x}{4} - \frac{x^2}{16}, \end{aligned} \quad (11.31)$$

with $c(\kappa, x)$ as in Equation (11.5). In particular,

$$\frac{\partial_x W_{\kappa,\mu}(x)}{W_{\kappa,\mu}(x)} = -\frac{\mu}{x} + \frac{1}{2x} + \frac{\kappa/2 - x/8}{\mu} + \frac{\kappa/4 - x/8}{\mu^2} + O(\mu^{-3}). \quad (11.32)$$

This gives

$$\frac{Q_{\kappa,h,\mu}(x)}{-2\mu W_{\kappa,\mu}(x)} = 1 + \frac{h}{2\mu} - \frac{\kappa x/2 - x^2/8}{\mu^2} + O(\mu^{-3}). \quad (11.33)$$

The target, in the same order variable, satisfies

$$\log \frac{\xi(2\mu + 1/2)}{2\pi^{-1/4} \Gamma(\mu) \pi^{-\mu} \mu^{9/4}} = -\frac{3}{32\mu} - \frac{9}{128\mu^2} + O(\mu^{-3}). \quad (11.34)$$

Thus the finite Robin parameter can cancel the first inverse-order mismatch after the power has been matched. It cannot be omitted from the comparison.

Theorem 11.7 (Constant Robin and spectral-shift exclusion). *No strictly positive realization of Equation (11.23) with real constant Robin condition $f'(u_0) = hf(u_0)$ has normalized inverse determinant F . This holds for every $c > 0$ and fixed real κ, u_0, h, λ , with the kinetic coefficient and exponential rate displayed in that equation.*

Proof. For $\lambda = 0$, write $D_R(w) = \mathcal{H}_R(w)/\mathcal{H}_R(0)$ in the positive realization. Equations (11.31)–(11.34) give the full real-axis comparison

$$\begin{aligned} \log \frac{F(-4\mu^2)}{D_R(-4\mu^2)} &= \log C_R + \mu \log \frac{x}{4\pi} + (5/4 - \kappa) \log \mu \\ &\quad + \frac{\delta_\kappa(x) - h/2}{\mu} + \frac{b_{\kappa,h}(x)}{\mu^2} + O(\mu^{-3}), \end{aligned} \quad (11.35)$$

$$C_R = -\frac{2\pi^{1/4} Q_{\kappa,h,0}(x)}{\sqrt{x} \xi(1/2)} > 0,$$

$$b_{\kappa,h}(x) = -\frac{9}{128} + \frac{\kappa^3}{6} - \frac{\kappa}{24} + \frac{\kappa x}{4} - \frac{x^2}{16} + \frac{h^2}{8}. \quad (11.36)$$

For a nonzero shift, the denominator is $\mathcal{H}_R(-\lambda)$ and the order is μ_λ from Equation (11.27). The extra order factor in Equation (11.33) only changes κ to $\kappa+1$ in the algebraic prefactor of Equation (11.28); it does not change its logarithmic $1/\mu$ term. Consequently an identity first forces $x = 4\pi$ and

$\kappa = 5/4$, then the leading normalization, and then $\lambda = 0$ from the coefficient of $(\log \mu)/\mu$. This argument uses the shifted normalization even when the unshifted operator is not invertible.

We may therefore use Equation (11.35). Its $1/\mu$ coefficient forces

$$h = h_* := 2\delta_{5/4}(4\pi) = 2\pi^2 - 5\pi + \frac{11}{8}. \quad (11.37)$$

Even granting $C_R = 1$, the remaining $1/\mu^2$ coefficient is

$$\begin{aligned} B_* &:= b_{5/4, h_*}(4\pi) = \frac{13}{64} + \frac{5\pi}{4} - \pi^2 + \frac{h_*^2}{8} \\ &= \frac{x^4 - 20x^3 + 90x^2 - 60x + 225}{512} \Big|_{x=4\pi} < 0. \end{aligned} \quad (11.38)$$

For an explicit rational sign bound, $157/50 < \pi < 22/7$ and monotonicity of $2p^2 - 5p + 11/8$ on that interval give

$$B_* < \frac{13}{64} + \frac{5}{4} \frac{22}{7} - \left(\frac{157}{50}\right)^2 + \frac{1}{8} \left(\frac{2123}{392}\right)^2 = -\frac{1583907247}{768320000} < 0. \quad (11.39)$$

Thus the logarithm cannot vanish identically. No numerical Whittaker evaluation is a premise of the exclusion. $\square$

The parameter tuple that survives the earlier filters is genuinely admissible, rather than excluded by a hidden positivity failure. Under the stated *left-endpoint* convention $f'(u_0) = hf(u_0)$, the quadratic form is

$$a_h[f] = \int_{u_0}^{\infty} (|f'|^2 + V|f|^2) du + h|f(u_0)|^2. \quad (11.40)$$

At $x = 4\pi$, $\kappa = 5/4$, and $h = h_* > 0$, the potential satisfies $V \geq 16\pi(\pi - 5/4) > 0$ and the boundary term is nonnegative. Hence no adverse trace-inequality penalty is needed for this convention. Numerically $h_* \approx 5.4062455342$ and $B_* \approx -2.0860522371$; Section 13.7 reports an independent normalization diagnostic. The failure is an exact second-order arithmetic mismatch after positivity, counting, scale, power, shift, and first-order matching have all survived.

Together, Corollary 11.6 and Theorem 11.7 cover every real constant separated self-adjoint left-endpoint condition for this Morse differential expression, including Neumann data as $h = 0$. The right endpoint is limit point. The extension to a positive kinetic coefficient and a common exponential rate is proved in Section 11.7; they do not leave a free spectral repair once the leading count is matched. Energy-dependent boundary data, additional potential terms, mode deletion, and a changed ratio of exponential rates remain outside this Morse-family exclusion.

11.7 Kinetic-rate normalization of the full Morse family

A positive kinetic coefficient and the common exponential rate do not leave an additional repair parameter when the ratio of the two rates remains 2 : 1. Consider the full differential expression

$$\mathcal{A} = -a \frac{d^2}{du^2} + c^2 e^{2\beta u} - 4c\kappa e^{\beta u} + \lambda, \quad u \geq u_0, \quad a, c, \beta > 0, \quad \kappa, u_0, \lambda \in \mathbb{R}. \quad (11.41)$$

The coordinate dilation and its unitary normalization are

$$v = \frac{u}{\sqrt{a}}, \quad (\mathcal{U}_a f)(v) = a^{1/4} f(\sqrt{a} v), \quad b := \beta \sqrt{a}, \quad x := ce^{\beta u_0}. \quad (11.42)$$

Here $\mathcal{U}_a : L^2([u_0, \infty), du) \rightarrow L^2([u_0/\sqrt{a}, \infty), dv)$ is unitary; no eigenvalue rescaling has been made in this first step.

Corollary 11.8 (Full kinetic-rate Morse exclusion). *No strictly positive realization of Equation (11.41), with a fixed real separated self-adjoint left-endpoint condition, has $\det(\mathbf{I} - w\mathcal{A}^{-1}) = F(w)$. This includes arbitrary positive kinetic coefficient, common rate, potential scale, endpoint, real κ , constant spectral shift, and real constant Robin parameter.*

Proof. Conjugation by $\mathcal{U}_a$ makes the kinetic coefficient one and replaces β by b . A second change $r = bv + \log(2c/b)$, with unitary factor $b^{-1/2}$, gives

$$\mathcal{A} \simeq b^2 \left(-\frac{d^2}{dr^2} + \frac{1}{4}e^{2r} - \frac{2\kappa}{b}e^r \right) + \lambda, \quad r \geq r_0 := \log \frac{2x}{b}. \quad (11.43)$$

The first and second changes preserve the class of real constant separated boundary conditions. Apply the Morse counting law [16, Theorem 3.2] with its parameter $k = -2\kappa/b$ and energy $(E - \lambda)/b^2$. For each fixed parameter tuple,

$$N_{\mathcal{A}}(E) = \frac{\sqrt{E}}{b\pi} \left[\log \sqrt{E} + \log \frac{2}{x} - 1 \right] + O(1). \quad (11.44)$$

The fixed shift changes the displayed smooth terms only by $O(E^{-1/2} \log E)$, absorbed in this bounded remainder. The count also implies $\mathcal{A}^{-1}$ is trace class in the strictly positive case.

If its determinant were F , its complete eigenvalue count would satisfy Equation (9.1). The two leading coefficients therefore force, successively,

$$\frac{1}{2b\pi} = \frac{1}{4\pi}, \quad b = \beta\sqrt{a} = 2, \quad x = ce^{\beta u_0} = 4\pi. \quad (11.45)$$

When $b = 2$, the first conjugated expression is exactly Equation (11.23) on $v \geq u_0/\sqrt{a}$. Dirichlet data stay Dirichlet, while $f'(u_0) = hf(u_0)$ becomes

$$(\mathcal{U}_a f)'(u_0/\sqrt{a}) = h_{\text{red}}(\mathcal{U}_a f)(u_0/\sqrt{a}), \quad h_{\text{red}} = \sqrt{a}h. \quad (11.46)$$

Corollary 11.6 and Theorem 11.7 now exclude every such normalized candidate. Unitary equivalence preserves the determinant, completing the proof. $\square$

The first counting term fixes the product $\beta\sqrt{a}$, not a and β separately. The residual freedom with that product fixed is a coordinate dilation, not an untested spectral adjustment. In the Robin branch the last first-order match would read $\sqrt{a}h = h_*$; its second-order residual is still $B_* \neq 0$. The remaining exclusions of scope are changes of the 2 : 1 rate ratio, additional potential terms, mode deletion, and energy-dependent boundary data. These are different spectral families, not free parameters of Equation (11.41).

12 Finite positivity does not close the infinite problem

A finite matrix calculation can disprove a proposed positive representation if it produces a rigorously negative quadratic value. A positive calculation in finitely many degrees has a different logical status. The following construction makes the distinction explicit, including the possibility of off-line zeros arbitrarily far above the tested regime.

Theorem 12.1 (Finite-tower nonclosure). *For every integer $q \geq 0$ and every $0 < \eta < 1/2$, there exists a real entire function G of order $1/2$ with $G(0) = 1$ and $G(-x) > 0$ for $x \geq 0$, such that its logarithmic-moment matrices are positive definite for $0 \leq N \leq q$, while G has nonreal zeros. The associated even function $G(z^2)$ can have all its zeros in $|\Im z| < 1/2$. The construction can also preserve the first two zeta counting terms.*

Proof. Start with

$$G_0(w) = \frac{\sin(\pi\sqrt{w})}{\pi\sqrt{w}} = \prod_{n=1}^{\infty} (1 - w/n^2). \quad (12.1)$$

Its logarithmic moments are $m_k^{(0)} = \zeta(2k+2)$, represented by the positive measure $\sum_{n \geq 1} n^{-2} \delta_{n^{-2}}$. Every finite Hankel matrix is therefore positive definite by the infinite-support argument in Theorem 6.2.

For $T > 0$, put $a_T = (T + i\eta)^2$ and define

$$G_T(w) = G_0(w)(1 - w/a_T)(1 - w/\overline{a_T}). \quad (12.2)$$

It has nonreal zeros $a_T, \overline{a_T}$, remains real entire of order $1/2$, and satisfies $G_T(-x) = G_0(-x)|1 + x/a_T|^2 > 0$. Its logarithmic moments are

$$m_k^{(T)} = m_k^{(0)} + 2\Re(a_T^{-k-1}). \quad (12.3)$$

For $|a_T| \geq 1$, the perturbation of the $(q+1) \times (q+1)$ Hankel matrix has operator norm at most $2(q+1)/|a_T|$. Choose T large enough that this is less than the smallest eigenvalue of $H_q^{(0)}$. The perturbed H_q remains positive definite, as do all its leading principal submatrices. The extra zeros of $G_T(z^2)$ are $\pm(T + i\eta)$ and $\pm(T - i\eta)$; all other zeros are real.

For the final assertion, replace G_0 by D_{exp} . Its positive trace-class realization gives strictly positive finite moment matrices in exactly the same way. Multiplication by the same two factors adds only a fixed finite quartet in the z -plane and does not change either leading counting coefficient in Equation (10.4). $\square$

This theorem does not alter F and does not construct an arithmetic counterexample. In particular, G_T is not claimed to have the Euler product of ζ . It shows why a proof of the specific arithmetic identity cannot be replaced by a finite list of matrix signs, strip symmetry, real-axis positivity, and two counting terms. Those properties coexist in the construction with nonreal zeros.

The theorem also does not exclude finite computation from a proof accompanied by an independent global estimate. A rigorously justified bound reducing the entire problem to a finite verification would be additional mathematical information. What fails is the unqualified inference from a finite positive prefix to positivity at every degree.

12.1 Finite-to-infinite arithmetic moment transfer

The finite-rank obstruction concerns one fixed representation, not a refinement family. A sequence can increase its rank while every individual member remains finite. The following consequence of Theorem 6.2 makes the required limiting statement explicit; it is a positivity-transfer result, not an independent proof of the arithmetic limit.

Theorem 12.2 (Finite positive trace-moment transfer). *RH is equivalent to the existence of positive finite-dimensional matrices K_r such that, for every fixed $n \geq 0$,*

$$\lim_{r \rightarrow \infty} \text{Tr } K_r^{n+1} = m_n. \quad (12.4)$$

The matrices need not act on the same finite-dimensional space. No assumption of convergence uniform in n is required.

Proof. For a fixed polynomial $p(t) = \sum_{j=0}^N c_j t^j$, positivity and finite-dimensional spectral calculus give

$$\sum_{i,j=0}^N \bar{c}_i c_j \text{Tr } K_r^{i+j+1} = \text{Tr}(K_r p(K_r)^* p(K_r)) \geq 0. \quad (12.5)$$

There are only finitely many terms for this p . Equation (12.4) therefore gives $q_\xi(p) \geq 0$. Since p was arbitrary, Theorem 6.2 proves RH. Conversely, under RH the finite diagonal matrices with entries $\gamma_1^{-2}, \dots, \gamma_r^{-2}$ satisfy Equation (12.4) by absolute convergence of Equation (6.3). This conditional construction proves the equivalence; it is not an independent source construction. $\square$

In Equation (12.5), self-adjointness gives $p(K_r)^* = \sum_i \bar{c}_i K_r^i$, which explains the conjugated coefficients on the left. The necessary information is the arithmetic value of every limiting moment. Conservation or refinement of an unrelated source operator cannot replace Equation (12.4). The result nevertheless shows why finite conservative discretizations remain legitimate ingredients of a proof. No single mesh is required to contain the complete arithmetic representation.

Corollary 12.3 (Degree-wise error transfer). *For a fixed N , suppose $K_r \geq 0$ and*

$$\max_{0 \leq n \leq 2N} \left| \text{Tr } K_r^{n+1} - m_n \right| \leq \epsilon_{r,N}. \quad (12.6)$$

Then $\lambda_{\min}(H_N) \geq -(N+1)\epsilon_{r,N}$. If $\epsilon_{r,N} \rightarrow 0$ for every fixed N , RH follows.

Proof. The matrix $H_N^{(r)} = (\text{Tr } K_r^{i+j+1})$ is positive semidefinite by Equation (12.5). The entrywise bound implies $\|H_N - H_N^{(r)}\| \leq (N+1)\epsilon_{r,N}$. Apply this estimate to unit-vector quadratic forms and then take the limit. The final assertion is Theorem 6.2. $\square$

The bound in Equation (12.6) must be proved against the *actual* moments, not defined retrospectively from the candidate's computed values. At a finite r it supplies an error statement, not all-degree positivity. No such all-degree arithmetic error estimate has been obtained for the source-compatible matrices in Equation (8.10).

12.2 A common Hilbert space and the determinant limit

For an operator-limit route, specify isometric embeddings $\iota_r : \mathcal{H}_r \rightarrow \mathcal{H}$ into a common Hilbert space and write $\widetilde{K}_r = \iota_r K_r \iota_r^*$. The embeddings, not just the matrix entries, are part of the realization. Assume

$$\|\widetilde{K}_r - K\|_1 \rightarrow 0, \quad (12.7)$$

where $\|\cdot\|_1$ is the trace norm. Completeness of the trace class and operator-norm convergence imply that K is positive trace class. If R bounds $\|\widetilde{K}_r\|$ and $\|K\|$, telescoping the powers gives

$$\left| \text{Tr } \widetilde{K}_r^{n+1} - \text{Tr } K^{n+1} \right| \leq (n+1)R^n \|\widetilde{K}_r - K\|_1. \quad (12.8)$$

Indeed, write the difference of powers as the sum of $n + 1$ products containing one factor $\widetilde{K}_r - K$ and use the trace-ideal inequality. Fredholm determinants also converge locally uniformly under Equation (12.7) [13, 25]. Neither assertion identifies the limit with F unless the arithmetic moments or prime trace are independently matched.

Operator norm alone does not suffice. If P_r projects onto the first r coordinates of ℓ^2 and $K_r = r^{-1}P_r$, then

$$\|K_r\| \rightarrow 0, \quad \text{Tr } K_r = 1, \quad \det(\text{I} - wK_r) = (1 - w/r)^r \rightarrow e^{-w}. \quad (12.9)$$

The determinant of the operator-norm limit is one, not e^{-w} . The example locates the failure in unresolved trace mass rather than in any individual large eigenvalue.

12.3 A sufficient trace-tightness estimate

A source approximation may first establish operator-norm convergence. The following lemma states an additional tail condition sufficient to upgrade it to the topology needed above. It uses only positive-operator block decomposition and Hilbert–Schmidt inequalities [13].

Lemma 12.4 (Uniform trace tightness). *Suppose $T_r \geq 0$ are trace-class operators on one Hilbert space, $\|T_r - T\| \rightarrow 0$, and $\sup_r \text{Tr } T_r \leq M < \infty$. Let P_m be finite-rank orthogonal projections, strongly increasing to I , and put $Q_m = \text{I} - P_m$. If*

$$\sup_r \text{Tr}(Q_m T_r Q_m) \leq \varepsilon_m, \quad \varepsilon_m \downarrow 0, \quad (12.10)$$

then T is positive trace class and

$$\|T_r - T\|_1 \leq (\text{rank } P_m)\|T_r - T\| + 2\varepsilon_m + 4\sqrt{M\varepsilon_m}. \quad (12.11)$$

In particular, $\|T_r - T\|_1 \rightarrow 0$.

Proof. Positivity passes to the operator-norm limit. For any finite-rank projection P , $\text{Tr}(PTP) = \lim_r \text{Tr}(PT_r P) \leq M$. Taking the supremum over such projections shows that T is trace class with $\text{Tr } T \leq M$. The same argument on the range of Q_m gives $\text{Tr}(Q_m T Q_m) \leq \varepsilon_m$.

For any positive trace-class A with $\text{Tr } A \leq M$ and $e = \text{Tr}(Q_m A Q_m)$, the block decomposition gives

$$\|A - P_m A P_m\|_1 \leq e + 2\sqrt{M e}. \quad (12.12)$$

The diagonal tail has trace norm e . For the off-diagonal block, factor $P_m A Q_m = (P_m A^{1/2})(A^{1/2} Q_m)$ and bound its trace norm by the product of the Hilbert–Schmidt norms. Apply Equation (12.12) to both T_r and T ; bound the remaining finite block by $(\text{rank } P_m)\|T_r - T\|$. This proves Equation (12.11). First choose m to control the tails and then r to control the finite block. $\square$

The projections in Equation (12.10) belong to the common Hilbert realization. A tail bound on a different, independently chosen basis at every refinement does not supply this hypothesis. The geometric preflights do not establish it, and a small conservation residual is not a substitute. The source-domain, excision, interface, and spectral limits must retain the declared order [20, 21].

12.4 Classical sign counts and delayed finite detection

Chebotarev's complement gives a useful classical comparison to finite-prefix nonclosure. For the order-one real entire-function setting of Section 6.3, let

$$\Delta_{-1}(f) = \Delta_0(f) = 1, \quad \Delta_n(f) = \det \left(\sum_{\ell} z_{\ell}^{-i-j-2} \right)_{i,j=0}^{n-1} \quad (n \geq 1), \quad (12.13)$$

where the nonzero zeros z_{ℓ} are counted with multiplicity. The formulation in [27, Theorem 10(ii)] states that, if the complete sequence $\{\Delta_{n-1}(f)\Delta_n(f)\}_{n \geq 0}$ contains exactly a finite number m of negative terms, the function has m *distinct* nonreal conjugate pairs and infinitely many real zeros. This is a global sign-count statement, not an assertion that a finite positive prefix determines the total. We use it only for classical context in the nondegenerate sign-count setting; vanishing leading determinants require separate treatment and are not assigned a sign through rounding. No root-simplicity or determinant-nondegeneracy hypothesis is added to Theorem 6.2.

For the construction of Theorem 12.1, the order-one comparison function is $f_T(z) = G_T(z^2)$, not the order-1/2 function G_T itself. The construction can delay failure in *both* moment towers. For either stated positive base G_0 , the shifted matrices are positive definite, since they are represented by the positive measure with weights λ_j^2 at its infinitely many distinct inverse zeros $\lambda_j > 0$. Equation (12.3) gives, for $|a_T| \geq 1$,

$$\|H_q^{(T)} - H_q^{(0)}\| \leq \frac{2(q+1)}{|a_T|}, \quad \|H_q^{(+),(T)} - H_q^{(+),(0)}\| \leq \frac{2(q+1)}{|a_T|^2}. \quad (12.14)$$

Choosing T sufficiently large makes both perturbations smaller than the corresponding least eigenvalues. The even/odd block identity then gives $\Delta_n(f_T) > 0$ for every $n \leq 2q+2$, although f_T has the nonreal quartet $\pm(T+i\eta)$ and $\pm(T-i\eta)$. That quartet comprises two distinct conjugate pairs in the z -plane, whereas G_T has one conjugate pair in the w -plane. Thus the first loss of strict positivity in the full lifted tower can be postponed past any prescribed finite prefix. Neither an eventual first negative index nor an exact global Chebotarev count is inferred from the truncated calculations; a zero determinant must also be distinguished from a negative one.

13 Reproducible numerical diagnostics

13.1 Computing without zero ordinates

The coefficients used here are computed directly from derivatives at the central point:

$$a_k = \frac{\xi^{(k)}(1/2)}{k!}, \quad f_j = (-1)^j \frac{a_{2j}}{a_0}, \quad (13.1)$$

followed by Equation (5.9). No table of zeta zeros is supplied to the calculation. This prevents the numerical input from presupposing the real spectral data that the programme is trying to derive.

Calculations used Python 3.13.5 and `mpmath` 1.3.0, with independent runs at 40 and 60 decimal working precision. The displayed digits agree between those runs. The software provides arbitrary-precision arithmetic, but these calculations use its ordinary floating-point context rather than interval enclosures [19]. The values are therefore reproducible numerical diagnostics, not certified sign proofs.

Table 1: Logarithmic moments from central derivatives. Values are rounded, non-interval-certified diagnostics.

n	m_n
0	$2.31049931154190 \times 10^{-2}$
1	$3.71725992852697 \times 10^{-5}$
2	$1.44173931400973 \times 10^{-7}$
3	$6.63031680252991 \times 10^{-10}$
4	$3.21366415061660 \times 10^{-12}$
5	$1.58769345149102 \times 10^{-14}$
6	$7.90332572404674 \times 10^{-17}$
7	$3.94647139999885 \times 10^{-19}$
8	$1.97326507374497 \times 10^{-21}$
9	$9.87216538329716 \times 10^{-24}$

To improve the scale of the finite matrices, put $b_n = 100^{n+1}m_n$ and $\tilde{H}_N = (b_{i+j})$. If $D_N = \text{diag}(1, 100, \dots, 100^N)$, then

$$\tilde{H}_N = 100D_NH_ND_N, \quad \det \tilde{H}_N = 100^{(N+1)^2} \det H_N. \quad (13.2)$$

This positive congruence preserves all sign questions. It is numerical conditioning, not an additional arithmetic or physical parameter.

Table 2: Leading determinants after the exact positive scalings in Equations (13.2) and (13.22). The prefixes are compatible with positivity; they do not certify either infinite tower. Both columns are evaluated through $N = 4$.

N	$\det \tilde{H}_N$	$\det \tilde{H}_N^{(+)}$
0	2.31049931154190	$3.71725992852697 \times 10^{-1}$
1	$1.94933555481914 \times 10^{-1}$	$3.86048846787094 \times 10^{-3}$
2	$2.17281051052625 \times 10^{-4}$	$3.11996028999615 \times 10^{-7}$
3	$1.17758171601017 \times 10^{-9}$	$9.24213895905459 \times 10^{-14}$
4	$1.96218205320998 \times 10^{-17}$	$7.86624627750459 \times 10^{-23}$

The rapid decrease of the determinants makes precision control important. Agreement between two precision settings is a practical diagnostic of stability, not a substitute for an enclosure of truncation and roundoff errors. A formal numerical certificate would need validated derivatives or validated theta integrals, interval propagation through Equation (5.9), and certified matrix positivity, for example through an interval LDL^* factorization. No claim of such certification is made here.

13.2 Independent checks of the failed candidate

The exponential asymptotic can also be inspected directly. Let c_{exp} be the positive constant in Equation (10.8). The quotient

$$R_{\text{exp}}(s) = \frac{F(-(s-1/2)^2)}{c_{\text{exp}}s^{9/4}D_{\text{exp}}(-(s-1/2)^2)} \quad (13.3)$$

has numerical values approximately 2.6975168723, 1.2678702215, and 1.0597685216 at $s = 20, 80, 320$, respectively. Their approach toward one is a diagnostic of the derived asymptotic, not its proof. The proof is Equation (10.9) and the gamma-ratio calculation.

For the Morse repair, $\delta_{9/4} \approx -1.8300625401$. The quantity $\mu(F/D_{9/4}/C_{9/4} - 1)$, evaluated at $s = 2\mu + 1/2$, is approximately -1.63031121 , -1.76862626 , and -1.80943692 for $\mu = 30, 100, 300$. These values agree with the limiting coefficient in Equation (11.6). Again, the family-level exclusion depends on the analytic nonzero coefficient, not on the floating-point values.

13.3 Quantitative prime-trace comparison

A numerical comparison becomes an exclusion test only after its errors are bounded. Write, for real $s > 1$ and $t = s - 1/2$,

$$\mathcal{T}_K(s) = 2t \operatorname{Tr}[K(I + t^2 K)^{-1}], \quad (13.4)$$

$$B_\xi(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi_\Gamma(s/2),$$

$$\mathcal{A}_\xi(s) = B_\xi(s) - \sum_{n=2}^{\infty} \Lambda(n) n^{-s}. \quad (13.5)$$

The target is $\mathcal{T}_K = \mathcal{A}_\xi$, not the prime sum without its completion factors. This comparison uses the convergent half-plane. The critical strip is $0 < \Re s < 1$ and the critical line is $\Re s = 1/2$; analytic continuation is well defined there, while the original absolutely convergent Euler product is not. The choice $s > 1$ controls this series manipulation, not a supposed instability of complex analysis [2, 17].

Proposition 13.1 (Positive resolvent-trace bound). *For positive trace-class K, Q on the same Hilbert space and real $s > 1$,*

$$|\mathcal{T}_K(s) - \mathcal{T}_Q(s)| \leq 2(s - \tfrac{1}{2}) \|K - Q\|_1. \quad (13.6)$$

No commutation between K and Q is required.

Proof. Put $a = t^2$. Multiplication on the left by $I + aK$ and on the right by $I + aQ$ verifies

$$K(I + aK)^{-1} - Q(I + aQ)^{-1} = (I + aK)^{-1}(K - Q)(I + aQ)^{-1}. \quad (13.7)$$

Both resolvents have operator norm at most one by positivity. The trace-ideal inequality bounds the trace norm of the right side by $\|K - Q\|_1$. Apply $|\operatorname{Tr} R| \leq \|R\|_1$ and multiply by $2t$. $\square$

Lemma 13.2 (Unconditional prime-power tail). *For an integer $P \geq 3$ and real $s > 1$,*

$$0 \leq \sum_{n>P} \frac{\Lambda(n)}{n^s} \leq \mathcal{R}_{\text{pr}}(P, s) := P^{1-s} \left(\frac{\log P}{s-1} + \frac{1}{(s-1)^2} \right). \quad (13.8)$$

Proof. The defining prime-power weights satisfy $0 \leq \Lambda(n) \leq \log n$. The function $x \mapsto (\log x)x^{-s}$ is decreasing for $x \geq 3$. Bound the sum by $\int_P^\infty (\log x)x^{-s} dx$ and integrate. No zero-location assumption enters. $\square$

Suppose a positive trace-class limit K has a declared positive approximation $\widetilde{K}_r$ with a proved bound $\|\widetilde{K}_r - K\|_1 \leq \eta_r$. Define the exact finite discrepancy

$$\Delta_{r,P}(s) = \mathcal{T}_{\widetilde{K}_r}(s) - \left[B_\xi(s) - \sum_{n=2}^P \Lambda(n) n^{-s} \right]. \quad (13.9)$$

Let $\widehat{\Delta}_{r,P}(s)$ be its computed value with certified absolute evaluation error at most ϵ_{eval} . Matrix assembly, quadrature, and source regularization must either be included in η_r or bounded separately in the evaluation error; no contribution is silently set to zero.

Corollary 13.3 (Certified rejection and limiting closure). *If, at one real $s > 1$,*

$$|\widehat{\Delta}_{r,P}(s)| > 2(s - \tfrac{1}{2})\eta_r + \mathcal{R}_{\text{pr}}(P, s) + \epsilon_{\text{eval}}, \quad (13.10)$$

then K does not satisfy Equation (7.6). Conversely, fix $1 < s_- < s_+$ and suppose $\eta_r \rightarrow 0$, $P_r \rightarrow \infty$, the evaluation error tends uniformly to zero, and

$$\sup_{s \in [s_-, s_+]} |\widehat{\Delta}_{r,P_r}(s)| \rightarrow 0. \quad (13.11)$$

Then K satisfies Equation (7.6) on the interior of that interval and RH follows.

Proof. Proposition 13.1 and Lemma 13.2 give

$$\left| \mathcal{T}_K(s) - \mathcal{A}_\xi(s) - \widehat{\Delta}_{r,P}(s) \right| \leq 2(s - \tfrac{1}{2})\eta_r + \mathcal{R}_{\text{pr}}(P, s) + \epsilon_{\text{eval}}. \quad (13.12)$$

The reverse triangle inequality proves rejection. On the compact interval, the first term is at most $2(s_+ - 1/2)\eta_r$ and the prime tail is at most $\mathcal{R}_{\text{pr}}(P, s_-)$. The limiting assumptions therefore imply equality throughout the interval. Apply Theorem 7.2. $\square$

A finite grid of apparently small discrepancies does not establish Equation (13.11); it requires an independently proved bound between grid points. Likewise, a computable prime tail does not provide the geometric trace-norm error. The latter remains a substantive spectral convergence problem for the candidate source construction.

13.4 Exclusion of a bounded operator family

The certified-cover requirement of the conservative paper can be specialized to arithmetic discrepancy [20]. Let Θ be a compact metric parameter set, with a certified finite δ -net $\{\theta_j\}_{j=1}^J$, and suppose positive trace-class candidates satisfy

$$\|K_\theta - K_{\theta'}\|_1 \leq L d(\theta, \theta') \quad (13.13)$$

for a proved uniform constant L . Fix one $s > 1$. Let $\widehat{\Delta}_j$ approximate $\mathcal{T}_{K_{\theta_j}}(s) - \mathcal{A}_\xi(s)$ with certified total error E_j , obtained for example from Equation (13.12). Then

$$\min_j (|\widehat{\Delta}_j| - E_j) - 2(s - \tfrac{1}{2})L\delta > 0 \quad (13.14)$$

excludes every K_θ in the declared family. Indeed, choose a net point within distance δ of θ and use Equation (13.6) to bound the change in discrepancy. Neither compactness alone nor a finite sample of parameters supplies L or a certified cover. Changing the family or its boundary law after rejection defines a new construction rather than repairing the rejected identity.

13.5 Exact manufactured algebraic diagnostics

A small rational example verifies the elimination, normalization, and trace algebra without claiming to realize the source geometry. In the interior–boundary order, take

$$\mathbf{C}_* = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}, \quad M_{b,*} = \mathbf{I}, \quad \mathbf{S}_* = \begin{pmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{pmatrix}, \quad K_* = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}. \quad (13.15)$$

The full matrix has positive leading minors 2, 3, 4. The Schur response has eigenvalues 1, 2, and K_* has eigenvalues 1, 1/2. Consequently

$$\mathrm{Tr} K_*^{n+1} = 1 + 2^{-n-1}, \quad \det H_1^{(*)} = \frac{1}{8}, \quad \det H_2^{(*)} = 0. \quad (13.16)$$

The rank-two termination is an exact control for Theorem 8.4, not a value of an arithmetic Hankel determinant. The source and static boundary determinants are visibly different:

$$\frac{\det(\mathbf{C}_* - w\mathbf{I})}{\det \mathbf{C}_*} = \frac{(2-w)((2-w)^2 - 2)}{4}, \quad \det(\mathbf{I} - wK_*) = (1-w)(1-w/2). \quad (13.17)$$

This directly verifies the need to distinguish them in Equation (8.13).

The accompanying `boundary_trace_diagnostics.py` uses exact rational arithmetic to check the Schur complement, minimizing extension, inverse normalization, finite moments, rank termination, and spectral-pencil factorization, together with the noncommuting resolvent identity at several rational parameters. Appendix D gives a minimal implementation. These checks verify algebra and code only. They are not a four-dimensional solver, a constructed singular-boundary functional, a trace-norm convergence certificate, or evidence for RH. The original numerical moment and Morse diagnostics remain unchanged.

13.6 Central normalization and the quarter-shift check

The familiar zero sum at $s = 1$ is related to, but not equal to, the central logarithmic moment m_0 . The Laurent expansion of ζ at one and the gamma values in Equation (3.2) give [2, 17]

$$\frac{\xi'(1)}{\xi(1)} = 1 + \frac{\gamma_E}{2} - \frac{1}{2} \log(4\pi) = M(-\tfrac{1}{4}) = \frac{1}{2} \sum_{\rho} \frac{1}{\rho(1-\rho)}. \quad (13.18)$$

Indeed, $\rho(1-\rho) = w_{\rho} + 1/4$ and differentiation of $F(-(s-1/2)^2) = \xi(s)/\xi(1/2)$ gives $\xi'(s)/\xi(s) = 2(s-1/2)M(-(s-1/2)^2)$. The sum in Equation (13.18) includes all nontrivial zeros with multiplicity and converges absolutely. In contrast,

$$m_0 = M(0) = \frac{\xi''(1/2)}{2\xi(1/2)}, \quad m_0 - M(-\tfrac{1}{4}) = \frac{1}{4} \sum_j \frac{1}{w_j(w_j + 1/4)}. \quad (13.19)$$

These identities do not assume RH. Numerically, $M(-1/4) \approx 0.0230957089661210$ and $m_0 - M(-1/4) \approx 9.28414929794 \times 10^{-6}$, consistently with Table 1. The distinction is a shift of spectral argument, not a discrepancy between two evaluations of the same moment. The added `scale_endpoint_diagnostics.py` reproduces these values and checks the free-endpoint comparison coefficients; its numerical values, like the earlier special-function diagnostics, are not interval certificates.

13.7 Robin normalization and second-order diagnostics

At the parameter tuple forced by the counting, power, shift, and first inverse-order conditions, the Whittaker normalization can be evaluated directly. The following evaluation uses $\kappa = 5/4$, $x = 4\pi$, $\lambda = 0$, and $h = h_*$ from Equation (11.37). The derivative is computed both by differentiation of W and from $W_{\kappa,\nu}(x) = e^{-x/2}x^{\nu+1/2}U(\nu+1/2-\kappa, 1+2\nu, x)$ with $\partial_x U(a, b, x) = -aU(a+1, b+1, x)$ [17, Sections 13.3 and 13.14].

$$\begin{aligned} h_* &\approx 5.4062455342297510, \\ W_{5/4,0}(4\pi) &\approx 0.0422104279006116, \\ \partial_x W_{5/4,0}(4\pi) &\approx -0.0167498199338303, \\ Q_{5/4,h_*,0}(4\pi) &\approx -0.6913792552608059, \\ C_R &\approx 1.0446407439095589, \\ B_* &\approx -2.0860522370547263. \end{aligned} \tag{13.20}$$

The displayed special-function values are non-interval-certified diagnostics; the exclusion uses the exact rational sign bound in Equation (11.39), not the numerical value of C_R .

The normalization therefore also differs numerically from one, but no claim of a certified special-function enclosure is required. More decisively, the numerical values of

$$\mu^2 \log \left(\frac{F(-4\mu^2)}{C_R D_R(-4\mu^2)} \right) \tag{13.21}$$

are approximately -1.8759731882 , -2.0296884422 , -2.0679420328 , and -2.0806912775 at $\mu = 30, 100, 300, 1000$, respectively. They approach the analytically derived B_* with the $O(\mu^{-1})$ remainder implied by Equation (11.35). The script `robin_shift_diagnostics.py` repeats the calculation at declared working precision, compares the two derivative formulas, and verifies the coefficient polynomials and rational sign bound by exact arithmetic. The original numerical outputs remain separate and unchanged.

13.8 The shifted Stieltjes prefix and block diagnostics

The shifted column in Table 2 makes the Hamburger–Stieltjes comparison numerically explicit without changing the original unshifted results. With $b_n = 100^{n+1}m_n$ and the diagonal D_N from Equation (13.2), the corresponding normalization is

$$\tilde{H}_N^{(+)} = (b_{i+j+1})_{i,j=0}^N = 100^2 D_N H_N^{(+)} D_N, \quad \det \tilde{H}_N^{(+)} = 100^{(N+1)(N+2)} \det H_N^{(+)}. \tag{13.22}$$

The additional factor of 100 per matrix entry relative to the unshifted congruence is essential. Both transformations preserve the exact sign conditions.

The script `shifted_tower_diagnostics.py` recomputes the central derivatives and recurrence at 40 and 60 decimal working digits. It does not read a zero table or reuse rounded moments as numerical inputs. The four shifted determinants, for $N = 0, 1, 2, 3$, agree at every digit printed in Table 2. The same script compares the directly assembled scaled Grommer determinant with its even/odd block product for sizes 1 through 8. The companion script `completed_tower_diagnostics.py` uses central Taylor order $4N + 4 = 20$ at $N = 4$ to compute m_9 and complete both columns through degree four. Fresh runs at 40 and 60 decimal working digits agree on every displayed

moment and determinant digit; the direct even/odd block checks cover sizes 1 through 10. These are floating-point consistency tests, not interval certificates or evidence for all-degree positivity.

The separate standard-library script `kinetic_rate_diagnostics.py` performs exact-rational checks of the reduction coefficients in Equation (11.43), the Robin transport, and the scaling and block identities on manufactured finite data. These implementation checks are distinguished from the symbolic all-parameter proof in Corollary 11.8 and from the unresolved arithmetic realization.

14 Invariants, artifacts, assumptions, and identifications

The framework’s invariant language must be indexed by an actual comparison class. The following distinctions prevent a numerical convention or an attractive spectrum from acquiring an unsupported source interpretation [4, 6].

Table 3: Logical status of the principal structures.

Class	Structure	Meaning and boundary
Governing source architecture	$\mathcal{C}^4$, $\mathcal{S} = \text{Orb}(p_0)$, typed ECL, foliation–label–clock distinction	The starting architecture of the TCGS application; not inferred from the signs in Tables 1–2.
Arithmetic invariance	Zeros and multiplicities of the fixed entire function F ; exact moments m_n	Exact mathematical data. A representation preserving them is not thereby identical to the source ontology.
Selector invariance	Unitary transport in Equation (8.4)	Preserves the Gram form along declared equivalence arrows; not across arbitrary changes of potential or endpoint.
Coordinate convention	$w = z^2$, inverse spectral variable, half-density normalization	Must be retained consistently; changing variables is not a change in the arithmetic target.
Numerical artifact	Working precision, rounding, scale factor 100, finite matrix size	Influences computation, not the exact sign condition. The factor 100 is an explicit congruence.
Model assumption	Exponential potential, Dirichlet endpoint, Morse correction κ	Independently testable choices. Their calibration to a counting law is not an ECL derivation.
Unproved identification	$q_\xi(p) = \ L_\sigma J_\sigma p\ ^2$ for every polynomial	The sufficient all-degree bridge. Its existence has not been obtained from the source geometry.
Excluded identification	A zeta zero is automatically a member of $\mathcal{S}$	A zero of an entire function is not a singularity of that function. Its logarithmic derivative has a pole, which is still a different mathematical type from a source singularity.

Class	Structure	Meaning and boundary
Proved model exclusion	The complete inverse determinant is not F for any strictly positive real constant separated Morse realization in Equation (11.41).	Free scale/endpoint: Corollary 11.3; constant shift: Corollary 11.6; Robin: Theorem 11.7; kinetic/rate: Corollary 11.8. This is a family-specific exclusion, not a general spectral no-go theorem.

Source invariants and their mathematical representatives must also remain distinct. The prime-power coefficients $\Lambda(n)$ are not declared to be direct measurements of Counterspace. They enter as exact arithmetic data on the right side of the bridge identity. If a source representation is eventually obtained, the identification will be a theorem relating those data to the representation, not a renaming of primes as singularities.

The same care applies to positivity. Positive definiteness of H_N is a mathematical property of a finite matrix. Positive semidefiniteness of all H_N is an infinite arithmetic assertion equivalent to RH. Positivity of O_σ is a theorem about a closed readout operator. These statements are related only through a proved identity such as Equation (8.2); their shared adjective does not establish that identity.

15 Discussion

15.1 What the geometric attempt establishes

The analysis moves the TCGS application from a broad source-shadow analogy to a sharply typed realization problem. The arithmetic target is fixed independently through ξ . The positive geometry is fixed independently through operator domains, norms, and quadratic forms. The Gram and prime-trace theorems specify the exact equality required between them. The construction is therefore not allowed to succeed merely by displaying the critical half, a real spectrum, or a visually plausible distribution of levels.

The explicit corridor model demonstrates both the opportunity and the obstruction. Noncompact geometry supplies the logarithmic increase in accessible length needed for the zeta density. This explains concretely why a logarithmic counting term need not require a temporal source dimension or an ordinary compact spectrum. Yet the fully computed determinant fails. A first repair within the Morse family removes the leading power defect only at one parameter and then fails at the next coefficient. A constant Robin endpoint supplies an additional power and one further matching coefficient, so it requires a separate calculation: even after that matching and exclusion of constant shifts, Equation (11.38) leaves a nonzero second-order term. The failure is not attributed to insufficient numerical precision; it is a proved mismatch of analytic functions.

The bounded exclusion is useful for subsequent source-geometry work. A candidate in the full Morse family of Equation (11.41), with any real constant separated endpoint condition, cannot satisfy Equation (7.6), however its parameters are interpreted. A different candidate must change the spectral realization in a mathematically substantive way. Possible changes include boundary data beyond a real constant separated condition, a nonlocal constitutive reduction, singular interface data, or a different operator class. These are design directions, not derived solutions, and each must reproduce the exact arithmetic identity rather than only the two counting coefficients.

15.2 The distinction between a criterion and a solution

An equivalence theorem does not establish either side of the equivalence unconditionally. Theorem 6.2 says that universal positivity settles RH; it does not prove universal positivity. Theorem 7.2 says that a positive realization of the prime trace settles RH; it does not produce that realization. Their value is to identify a more explicit mathematical target, not to relabel the original assertion as a completed proof.

This distinction is particularly important for a framework with a complete source ontology. Completeness of source content is not the same proposition as completeness of a chosen mathematical invariant or faithfulness of a selected representation. The framework itself distinguishes the full profile from one readout and an observability operator from a Hamiltonian [4, 6]. The present analysis applies those distinctions to arithmetic rather than weakening them at the point where a desired theorem becomes available.

The mathematical conclusions also do not depend on assigning physical ontology to the auxiliary gamma measures, Mellin coordinate, or heat parameter. Those objects are devices in analysis. The source framework constrains how a physical realization would be interpreted, while the proofs remain statements about explicitly defined functions and operators. This division avoids both an ontological inflation of mathematics and an exemption of framework-inspired mathematics from ordinary proof obligations.

15.3 A decisive next construction

The most direct constructive target is a positive trace-class K specified from an admissible source reduction, together with a derivation of Equation (7.6) in $s > 1$. This target has two advantages. It does not require prior knowledge of the zero ordinates, and its prime-power series is absolutely convergent. A successful derivation could use an independently defined arithmetic action, trace formula, or constitutive insertion map. None has been supplied by the constructions in this paper.

A second target is an all-degree proof that the logarithmic transform of the exact theta moments satisfies Equation (8.2). The raw theta kernel is known explicitly, and the coefficient recurrence is executable. The missing result is a positive representation of the nonlinear logarithmic moment transform. Establishing positivity of the raw theta density again would not address this remaining problem.

The boundary conditions for success are exact. The operator must be positive on its stated domain; the trace must be finite where used; multiplicities and the full determinant must agree; selector changes claimed to be equivalent must preserve the construction; and no unknown arithmetic fact may be inserted as a spectral assumption. A finite positive prefix, an asymptotic density match, or an operator built from already selected real zeros does not meet these conditions.

15.4 Limitations of the present results

No constructive arithmetic realization of Info , $\text{Read}_{S,G}$, or J_σ has been established here. No positivity-preservation theorem for the full logarithmic moment transform has been proved. The numerical matrices are finite and non-interval-certified. The compact-elliptic obstruction applies only to the operator class stated in Proposition 9.1, and the Morse exclusion to the full constant-coefficient differential expression in Equation (11.41), with rate ratio 2 : 1, arbitrary real constant spectral shift, and fixed real separated left-endpoint condition, as proved in Corollary 11.8 using Corollary 11.6 and Theorem 11.7. Neither is a general impossibility theorem for spectral approaches.

The foundational and application papers on gravity, biology, quantum readout, and consciousness provide the source architecture and its type discipline. Their domain-specific empirical claims are not used as premises for an arithmetic theorem. The three constituent framework works cited here are those needed for source geometry, typed Counterduction, and operator realization. This restriction keeps the mathematical dependence explicit without treating unrelated successes or analogies as evidence for RH.

Finally, the moment formulation is grounded in the classical Grommer–Hamburger real-zero method, with its reciprocal-square normalization distinguished explicitly in Section 6.3 [11, 12, 27]. The Bessel–Morse spectral family has established antecedents [16]. The paper’s results should be evaluated as an integrated constructive investigation with explicit transfer conditions and exclusions, not as a claim to have invented moment theory, Fredholm determinants, or spectral interpretations of zeta functions.

15.5 What the conservative boundary construction adds

The conservative and reciprocal-foam constructions change the available geometric starting point without supplying the missing arithmetic identity. Compatible cochains give an explicit positive frozen reciprocal sector. A shared boundary functional, once derived, would place the singular channels in the same variational assembly. Positive interior elimination then produces a boundary response and supplies a naturality theorem under the stated selector transports. These are constructive statements at a declared mathematical level; they do not turn S_r or K_r into the ontic ECL [20, 21].

Two distinctions now prevent a misleading success claim. First, the static boundary determinant need not equal the bulk determinant: Equation (8.13) displays the missing interior factor and spectral dependence. Second, a smooth compact boundary response remains subject to a Weyl obstruction, even though it is nonlocal. The proposed use of interfaces is therefore a method of generating and testing candidates, not a declaration that a boundary construction automatically possesses the arithmetic density.

The finite-to-infinite result addresses a different gap. Theorem 12.2 shows that finite matrices can participate in a proof when their traces converge to every actual arithmetic moment. Lemma 12.4 gives sufficient spectral-tail control for a trace-class operator limit, while Equation (13.12) separates geometric approximation, prime truncation, and numerical evaluation. The resulting exclusions are deterministic mathematical comparisons; they do not require a physical laboratory experiment. A finite positive prefix still cannot replace the stated limits.

The strongest remaining construction is an independently specified source representation whose boundary or bulk operator satisfies either Equation (12.4) or Equation (7.6). The geometric papers do not derive this arithmetic representation, their shared singular-boundary functional, or the trace-tightness bounds required here. Their liquid-lens, curvature, magnetic, and horizon sectors retain their own purposes and limits. They are not premises of the number-theoretic proof and are not imported as arithmetic evidence [20, 21].

The arithmetic content is specific. The prime-power term can be written

$$\sum_{n=2}^{\infty} \Lambda(n)n^{-s} = \sum_p \sum_{k=1}^{\infty} (\log p)e^{-sk \log p}, \quad s > 1. \quad (15.1)$$

A proposed corridor interpretation must account for these particular weights, repetitions, and lengths, as well as the completion terms in $B_{\xi}(s)$. General recurrence, self-similarity, or nontrivial

topology does not do so. Arithmetic inputs may legitimately be declared in a mathematical model; declaring them is not the same as deriving their source representation. The restriction is against hiding the desired zero-location conclusion in the operator definition.

15.6 Weil kernels as an arithmetic intermediate

The source-to-arithmetic construction need not proceed through an additional local potential. Appendix E identifies a continuous kernel whose exact transform is the logarithmic-moment function already used here. Theorem E.3 transfers an independently derived source Gram representation of that kernel to the existing RH criteria. Proposition E.4 also shows why the characteristic ratio proposed in a nonlocal extension programme must not be called the Fredholm determinant: its zeros are simple even when the corresponding zero of ξ has higher multiplicity, which remains encoded in the derivative.

The comparison gives additional mathematical tests, not an exemption from arithmetic identification. A shifted positive norm cannot establish positivity of its unshifted form; a signed first-order spectrum doubles the reciprocal-square tower unless the appropriate sector is selected; finite-window arithmetic must be combined with an all-window or complex-function limit. The resulting source-realization requirement remains consistent with the independently specified norms and selector transport in Section 8. No nonlocal limit or source insertion is assumed to exist merely because its conditional consequences can be proved.

16 Conclusion

The moment and spectral formulations of the Riemann hypothesis become directly comparable when their arithmetic and positivity requirements are separated and then connected by an exact identity. In the normalization used here,

$$\text{RH} \iff H_N \succeq 0 \quad \text{for all } N \iff F(w) = \det(I - wK), \quad K \geq 0 \text{ trace class.} \quad (16.1)$$

The prime-trace identity in Theorem 7.2 expresses the same target entirely within the absolutely convergent prime-power half-plane. The constitutive Gram-transfer theorem specifies what a TCGS arithmetic realization must preserve, and the finite-rank obstruction shows that no single finite protected sector can supply the complete representation.

The geometric construction has been carried far enough to be rejected precisely. A positive exponential corridor reproduces the leading arithmetic density but has the wrong determinant. A one-parameter Morse correction removes its power mismatch only at $\kappa = 9/4$, where a nonzero inverse-order coefficient still forbids equality. Corollary 11.3 extends the exclusion to free potential scale and endpoint: their only spectral combination is fixed to 4π by the second counting coefficient, so neither supplies a remaining repair parameter. These are mathematical results of the attempted construction, not evidence that all geometric routes fail.

The Morse-family exclusion now also permits arbitrary potential scale, endpoint, constant spectral shift, and real constant separated endpoint condition. Corollary 11.8 additionally permits positive kinetic coefficient and common exponential rate: the first counting term fixes their product $\beta\sqrt{a} = 2$, and the second fixes $ce^{\beta u_0} = 4\pi$. After the scale and power are fixed, the shift is forced to zero; the Robin parameter can remove the first-order term but cannot remove the explicit second-order residual. This conclusion uses neither a numerical estimate of the normalization constant nor an extrapolation from a finite list of eigenvalues.

The unresolved step is now explicit: derive, without assuming RH, a source-grounded positive representation of the full logarithmic moment form or the exact prime-weighted resolvent trace. No such derivation is established in this manuscript, and the Riemann hypothesis is therefore not claimed as solved. The result is a rigorous arithmetic realization programme with a proved criterion, an executed model family, and exact conditions separating a genuine proof from a spectral analogy.

The compatible-source extension supplies a further conditional route: positive interior elimination yields finite boundary candidates, coefficientwise convergence of their traces to the arithmetic moments implies RH, and trace tightness controls an operator limit when that route is used. The static/bulk determinant distinction and the regular boundary-power obstruction prevent artificial spectral closure. The rational checks establish the finite algebra exactly, whereas the shared source boundary law and the all-degree arithmetic identification remain unconstructed.

The continuous-kernel extension adds exact conversion formulas and conditional transfer mechanisms without changing this conclusion. Its finite-window arithmetic is explicit; its source Gram witnesses and nonlocal analytic limit remain unconstructed. The distinction among the moment determinant, the screw kernel, and the characteristic ratio prevents positivity or spectral agreement in one representation from being silently promoted to equality in another.

A Derivation of the Whittaker comparison coefficient

This appendix proves Lemma 11.1, including the term used to exclude the only power-matched Morse candidate. The integral representation [17], for fixed $x > 0$ and sufficiently large real μ , is

$$W_{\kappa,\mu}(x) = \frac{x^{\mu+1/2}2^{-2\mu}}{\Gamma(\mu+1/2-\kappa)} \int_1^\infty e^{-xt/2}(t-1)^{\mu-1/2-\kappa}(t+1)^{\mu-1/2+\kappa} dt. \quad (\text{A.1})$$

After $y = xt/2$, this becomes

$$W_{\kappa,\mu}(x) = \frac{x^{1/2-\mu}\Gamma(2\mu)}{\Gamma(\mu+1/2-\kappa)} \mathbb{E}\left[R_\mu(Y)\mathbf{1}_{\{Y>x/2\}}\right], \quad (\text{A.2})$$

where Y has gamma shape 2μ , unit scale, and

$$R_\mu(y) = \left(1 - \frac{x^2}{4y^2}\right)^{\mu-1/2} \left(\frac{1+x/(2y)}{1-x/(2y)}\right)^\kappa. \quad (\text{A.3})$$

The expectation is an auxiliary integral notation; it makes no claim about physical randomness.

On $y \geq \mu$ and for μ sufficiently large, Taylor expansion gives uniformly

$$R_\mu(y) = 1 + \frac{\kappa x}{y} - \frac{(\mu-1/2)x^2}{4y^2} + O(\mu^{-2}). \quad (\text{A.4})$$

Indeed, the logarithm has the two displayed terms with remainder $O(\mu/y^4 + y^{-3})$, while the square of its leading part is $O(\mu^{-2})$ on this region.

The omitted region contributes an exponentially small error. To see this without an uncontrolled expansion near $y = x/2$, put $v = x/(2y) \in (0, 1)$. If $\mu - 1/2 \geq |\kappa|$, the two factors $(1-v)^{\mu-1/2-\kappa}(1+v)^{\mu-1/2+\kappa}$ imply $R_\mu(y) \leq 2^{2|\kappa|}$. Moreover, for Y of shape 2μ ,

$$\Pr(Y \leq \mu) \leq e^\mu \mathbb{E}e^{-Y} = \left(\frac{e}{4}\right)^\mu.$$

The analogous negative-moment tails are exponentially small as well, by applying the same bound to gamma shapes $2\mu - 1$ and $2\mu - 2$ and multiplying by their gamma-ratio factors. Thus the full moments may replace the truncated moments in Equation (A.4).

Using $\mathbb{E}Y^{-1} = (2\mu - 1)^{-1}$ and $\mathbb{E}Y^{-2} = ((2\mu - 1)(2\mu - 2))^{-1}$, we obtain

$$\mathbb{E}\left[R_\mu(Y)\mathbf{1}_{\{Y>x/2\}}\right] = 1 + \frac{\kappa x/2 - x^2/16}{\mu} + O(\mu^{-2}). \quad (\text{A.5})$$

The duplication formula and the ordinary gamma-ratio expansion give

$$\begin{aligned} \frac{x^{1/2-\mu}\Gamma(2\mu)}{\Gamma(\mu+1/2-\kappa)} &= \frac{\sqrt{x}}{2\sqrt{\pi}}\Gamma(\mu)(x/4)^{-\mu}\frac{\Gamma(\mu+1/2)}{\Gamma(\mu+1/2-\kappa)} \\ &= \frac{\sqrt{x}}{2\sqrt{\pi}}\Gamma(\mu)(x/4)^{-\mu}\mu^\kappa\left[1 - \frac{\kappa^2}{2\mu} + O(\mu^{-2})\right]. \end{aligned} \quad (\text{A.6})$$

Multiplication by Equation (A.5) proves Equations (11.4)–(11.5). In particular, the coefficient excluding $\kappa = 9/4$ is not inferred from a numerical fit to Whittaker values.

A.1 Second-order coefficients and uniform differentiation

The next coefficient can be extracted from the same integral without fitting special-function values. Put $\alpha = \kappa x$, $\beta = x^2/4$, and let Y have the auxiliary gamma distribution of shape 2μ from Equation (A.2). Uniformly on $y \geq \mu$ and on each fixed compact positive x -interval,

$$R_\mu(y) = 1 + \frac{\alpha}{y} - \frac{(\mu - 1/2)\beta}{y^2} + \frac{1}{2}\left(\frac{\alpha}{y} - \frac{\mu\beta}{y^2}\right)^2 + O(\mu^{-3}). \quad (\text{A.7})$$

This follows by expanding $\log(1 - x^2/(4y^2))$ and $\log((1 + x/(2y))/(1 - x/(2y)))$ and then exponentiating. The omitted logarithmic terms are $O(\mu y^{-4} + y^{-3})$, and the cubic exponential terms are $O(\mu^{-3})$ on this region. The same estimates hold after one x -derivative.

For the complementary region $x/2 < y < \mu$, the earlier gamma tail bound remains valid. The derivative also has an integrable uniform estimate: with $v = x/(2y)$,

$$\partial_x R_\mu(y) = \frac{R_\mu(y)}{1 - v^2} \left(\frac{\kappa}{y} - \frac{(\mu - 1/2)x}{2y^2} \right). \quad (\text{A.8})$$

For $\mu - 3/2 \geq |\kappa|$, the first factor is bounded by $2^{2|\kappa|}$ by the same endpoint-factor argument used in Appendix A. The remaining factor is $O(1 + \mu)$ uniformly for x in a compact positive interval. The resulting tail and derivative tail are exponentially small. The moving lower endpoint contributes no term for sufficiently large μ , since $R_\mu(x/2) = 0$. Negative-moment tails are controlled by the corresponding shifted gamma shapes.

Using $\mathbb{E}Y^{-j} = \prod_{\ell=1}^j (2\mu - \ell)^{-1}$ for $j = 1, 2, 3, 4$, Equation (A.7) gives

$$\begin{aligned} e_1 &= \frac{\alpha}{2} - \frac{\beta}{4}, \\ \mathbb{E}\left[R_\mu(Y)\mathbf{1}_{\{Y>x/2\}}\right] &= 1 + \frac{e_1}{\mu} + \frac{(\alpha - \beta)/4 + e_1^2/2}{\mu^2} + O(\mu^{-3}), \\ \log \mathbb{E}\left[R_\mu(Y)\mathbf{1}_{\{Y>x/2\}}\right] &= \frac{\kappa x/2 - x^2/16}{\mu} + \frac{\kappa x/4 - x^2/16}{\mu^2} + O(\mu^{-3}). \end{aligned} \quad (\text{A.9})$$

The bounds just proved justify differentiation of this asymptotic with respect to x , rather than differentiation of an uncontrolled pointwise remainder.

For fixed a, b , the logarithmic gamma-ratio expansion is

$$\log \frac{\Gamma(\mu + a)}{\Gamma(\mu + b)} = (a - b) \log \mu + \frac{B_2(a) - B_2(b)}{2\mu} - \frac{B_3(a) - B_3(b)}{6\mu^2} + O(\mu^{-3}), \quad (\text{A.10})$$

where $B_2(t) = t^2 - t + 1/6$ and $B_3(t) = t^3 - 3t^2/2 + t/2$ [17, Section 5.11]. With $a = 1/2$, $b = 1/2 - \kappa$, this becomes

$$\log \frac{\Gamma(\mu + 1/2)}{\Gamma(\mu + 1/2 - \kappa)\mu^\kappa} = -\frac{\kappa^2}{2\mu} + \frac{-\kappa^3/6 + \kappa/24}{\mu^2} + O(\mu^{-3}). \quad (\text{A.11})$$

Combining with Equation (A.9) proves Equation (11.31). Its differentiated version gives Equation (11.32). Independently, substituting $a = 1/4$, $b = 0$ in Equation (A.10) gives $-3/(32\mu) - 1/(128\mu^2)$. Adding $\log(1 - 1/(16\mu^2)) = -1/(16\mu^2) + O(\mu^{-4})$ from the elementary factor in $\xi(2\mu + 1/2)$ gives $-9/(128\mu^2)$, proving Equation (11.34); the zeta correction is exponentially small.

Finally, taking the logarithm of Equation (11.33) contributes $h/(2\mu) + (-\kappa x/2 + x^2/8 - h^2/8)/\mu^2$ to the Robin characteristic. Subtraction from the target yields the coefficient $b_{\kappa, h}(x)$ in Equation (11.36). The nonzero residual in Equation (11.38) is therefore obtained from the same controlled integral and gamma expansions as the Dirichlet coefficient, not from a numerical estimate of C_R .

B A minimal reproducibility implementation

The following self-contained code reproduces the unshifted moment and determinant prefixes through m_8 . Its Taylor order is $4N + 2$, because H_N requires $m_0, \dots, m_{2N}$ and the recurrence requires $f_1, \dots, f_{2N+1}$. It contains no calls to routines that return zeta zeros. The separate source package includes a command-line implementation, precision-specific JSON results, and the Morse asymptotic diagnostic.

Listing 1: Finite logarithmic-moment diagnostics; not an interval certificate.

```

1 import mpmath as mp
2
3 # Run once at 40 and once at 60 decimal working precision.
4 mp.mp.dps = 60
5 N = 4
6 scale = mp.mpf(100)
7
8 def xi(s):
9     return (s * (s - 1) / 2 * mp.pi**(-s/2)
10            * mp.gamma(s/2) * mp.zeta(s))
11
12 a = mp.taylor(xi, mp.mpf('0.5'), 4*N + 2)
13 f = [(-1)**j * a[2*j] / a[0] for j in range(2*N + 2)]
14 m = []
15 for n in range(2*N + 1):
16     value = -(n + 1) * f[n + 1]
17     value -= mp.fsum(f[k] * m[n-k] for k in range(1, n+1))
18     m.append(value)
19
20 b = [scale**(n+1) * value for n, value in enumerate(m)]
21 for n, value in enumerate(m):
22     print('moment', n, mp.nstr(value, 25))
23 for degree in range(N + 1):

```

```

24     H = mp.matrix([[b[i+j] for j in range(degree+1)]
25                     for i in range(degree+1)])
26     print('determinant', degree, mp.nstr(mp.det(H), 25))
27
28     def morse_check(mu):
29         k = mp.mpf(9) / 4
30         x = 4 * mp.pi
31         s = 2 * mu + mp.mpf('0.5')
32         central = xi(mp.mpf('0.5'))
33         C = 2 * mp.whitw(k, 0, x) / (central * mp.pi**mp.mpf('.25'))
34         target = xi(s) / central
35         candidate = mp.whitw(k, mu, x) / mp.whitw(k, 0, x)
36         return mu * (target / candidate / C - 1)
37
38     for mu in (30, 100, 300):
39         print('Morse correction', mu, mp.nstr(morse_check(mp.mpf(mu)), 20))
40     print('Analytic limit', mp.nstr(mp.pi**2 - 9*mp.pi/2
41                                     + mp.mpf(39)/16, 20))

```

For rigorous numerical certification, each displayed scalar would need an enclosure rather than a precision-dependent approximation. Such an upgrade would strengthen the finite diagnostics only. Theorem 12.1 would still prevent the certified finite prefix from becoming an all-degree result without a separate global argument.

C Notation and proof-dependence summary

Table 4: Symbols whose formal types must remain distinct.

Symbol	Meaning
$\mathcal{C}^4, \Sigma^3$	Counterspace source representation and physical Shadow carrier.
Ξ, M_Ξ	Admissible source configuration and its informational organization.
$\mathcal{E}_S, \mathcal{E}_{S,\Xi}$	Constitutive map and complete profile value.
$\sigma, \mathcal{O}_S^\sigma$	Readout selector and physical-readout map.
L_σ, O_σ	Closed readout derivative and observability operator $L_\sigma^* L_\sigma$.
$\xi(s), \Xi(z)$	Riemann completion and its critical-line-centered entire function.
$F(w)$	Normalized even-power reduction; no square-root branch is chosen.
w_j, λ_j	Zeros of F ; under RH, inverse squared ordinates $\lambda_j = \gamma_j^{-2}$.
A_{2j}, f_j	Raw theta moments and Taylor coefficients of F .
m_n, H_N, q_ξ	Logarithmic moments, finite Hankel matrices, and their polynomial form.
J_σ	Unconstructed arithmetic insertion map in the Gram-transfer condition.
K, A	Positive trace-class arithmetic candidate and its inverse on the nonzero spectral subspace.
A_{exp}, A_κ	Explicit exponential and calibrated Dirichlet Morse test operators.
D_{exp}, D_κ	Their normalized Fredholm determinants; neither equals F .
$\mathcal{A}, a, \beta$	General Morse realization in Equation (11.41), its positive kinetic coefficient, and the lower exponential rate.
b, x, λ	Kinetic-rate invariant $b = \beta\sqrt{a}$, scale-endpoint combination $x = ce^{\beta u_0}$, and constant spectral shift.
h, h_{red}	Original left-endpoint Robin coefficient and its unitary transport $h_{\text{red}} = \sqrt{a} h$.
$\mathcal{H}_D, \mathcal{H}_R$	Unshifted Dirichlet and Robin characteristic functions in Equation (11.24); not yet normalized determinants.
$Q_{\kappa,h,\nu}(x)$	Robin characteristic combination $2x\partial_x W_{\kappa,\nu}(x) - (1+h)W_{\kappa,\nu}(x)$.
D_R, C_R	Normalized Robin determinant at $\lambda = 0$ and its asymptotic normalization constant; shifted normalization uses Equation (11.25).
h_*, B_*	The first-order matched Robin coefficient and the nonzero second-order obstruction in Equations (11.37) and (11.38).
$H_N^{(+)}, \tilde{H}_N^{(+)}$	Shifted Stieltjes moment matrix and its positively scaled numerical version.
$\Delta_n(f)$	Classical inverse-power Hankel determinant for an order-one function f ; it is not the diagonal numerical scaling D_N .
$\Lambda(n), \psi_\Gamma$	Von Mangoldt prime-power weights and gamma logarithmic derivative.
τ, μ, κ	Heat probe parameter, large Whittaker order, and Morse model parameter; none is an ontic temporal dimension.

The proof dependence can be read without the framework interpretation. No unconditional analytic or operator result in this paper uses the TCGS axioms as a proof premise. The framework supplies the realization setting and admissibility restrictions; the conditional transfer statements require the separately displayed mathematical hypotheses. Standard analytic properties of ξ give Lemma 3.1; the Hamburger theorem then gives Theorem 6.2. Positive trace-class theory gives Theorems 7.1 and 7.2. A *proved* Gram identification would invoke these through Theorem 8.2; that identification remains unconstructed. Independently, classical spectral theory and the explicit calculations give Propositions 9.1 and 9.2, the corridor construction, and Theorems 10.3 and 11.2. Corollary 11.3 uses the same Whittaker expansion and the scale–endpoint counting law. Lemma 11.4 and Proposition 11.5 justify the complex-plane product step; Corollary 11.6 and Theorem 11.7 use the displayed shift and second-order asymptotics, independently of numerical normalization values; Equations (6.9)–(6.12) relate the arithmetic indexing to the classical criterion. Corollary 11.8 uses a unitary dilation and the two leading counting terms to reduce the full Morse family to the already excluded one. Section 12.4 gives classical sign-count context and the two-tower finite-prefix comparison; the shifted numerical column uses Equation (13.22). The finite numerical results are not premises in any of these proofs.

D Exact boundary-response verification

The following independent rational check reproduces the inverse, moment sequence, and first rank-termination conditions in Equations (13.15)–(13.16), and verifies the noncommuting resolvent identity. It needs no external package. The complete command-line script additionally checks the minimizing extension, full characteristic polynomial, and spectral-pencil factorization, and writes `boundary_trace_diagnostics.json`. All 28 checks in that script pass using exact fractions. These manufactured checks do not evaluate the source boundary functional or the Riemann prime trace.

Listing 2: Exact rational verification of the manufactured boundary example.

```

1  from fractions import Fraction as F
2
3  I = [[F(1), F(0)], [F(0), F(1)]]
4  S = [[F(3, 2), F(-1, 2)], [F(-1, 2), F(3, 2)]]
5  K = [[F(3, 4), F(1, 4)], [F(1, 4), F(3, 4)]]
6  Q = [[F(3, 4), F(0)], [F(0), F(1, 4)]]
7
8  def mul(A, B):
9      return [[sum((A[i][k]*B[k][j] for k in range(2)), F(0))
10              for j in range(2)] for i in range(2)]
11
12  def add(A, B, c=F(1)):
13      return [[A[i][j] + c*B[i][j] for j in range(2)]
14              for i in range(2)]
15
16  def det(A):
17      return A[0][0]*A[1][1] - A[0][1]*A[1][0]
18
19  def inv(A):
20      d = det(A)
21      if not d:
22          raise ValueError('Singular matrix')
23      return [[A[1][1]/d, -A[0][1]/d],
24              [-A[1][0]/d, A[0][0]/d]]
25

```

```

26 def check(condition):
27     if not condition:
28         raise ArithmeticError('Exact identity failed')
29
30 check(mul(S, K) == I)
31 check(mul(K, Q) != mul(Q, K))
32 P, moments = I, []
33 for n in range(5):
34     P = mul(P, K)
35     moments.append(P[0][0] + P[1][1])
36     check(moments[-1] == 1 + F(1, 2**(n+1)))
37 a, b, c, d, e = moments
38 check(a*c - b*b == F(1, 8))
39 check(a*(c*e-d*d)-b*(b*e-c*d)+c*(b*d-c*c) == 0)
40 for t in (F(1), F(3, 2), F(4)):
41     RK, RQ = inv(add(I, K, t*t)), inv(add(I, Q, t*t))
42     left = add(mul(K, RK), mul(Q, RQ), F(-1))
43     right = mul(mul(RK, add(K, Q, F(-1))), RQ)
44     check(left == right)
45 print('Exact manufactured checks passed; no RH claim.')

```

E Weil positivity, screw kernels, and nonlocal spectral limits

This appendix connects the fixed arithmetic target of Sections 8 and 12 to a continuous prime-containing kernel and to recent nonlocal spectral constructions. Its normalization identities and transfer estimates are proved below. The external real-zero and screw-function criteria retain their own hypotheses; none supplies an unconstructed source insertion map. The governing source/readout distinction is the one in Section 2, not an identification of an arithmetic function with the ECL [4, 6].

E.1 Enumeration, arithmetic forms, and source realization

Prime-counting constructions such as López Subia's [34, 35] concern arithmetic enumeration, which is distinct from the positive-operator realization considered here.

For the present problem, any verified enumeration of primes can generate the coefficients $\Lambda(n)$ in Equation (15.1). Enumeration alone neither fixes a Hilbert norm nor proves the prime-trace identity. Indeed, a finite diagonal prime indicator is positive, but its only nonzero eigenvalue is 1; its determinant is $(1 - w)^r$, where r is the number of selected primes. It therefore cannot be the nonpolynomial function F . The distinction is algebraic: arithmetic data, a positive representation, and their exact identification are three different inputs. This elementary example also illustrates the rank obstruction in Section 8.

E.2 A continuous arithmetic function and the exact moment normalization

Let $\Phi_L(z, s, b) = \sum_{k \geq 0} z^k (k + b)^{-s}$ denote the Hurwitz–Lerch function on the arguments used below, and put $C_{1/4} = \zeta(2, 1/4)$. To avoid conflict with the source field Ψ , write Suzuki's arithmetic

function as Ψ_{ar} . For $u \geq 0$, its explicit definition is [29, Equation (1.1)]

$$\begin{aligned} \Psi_{\text{ar}}(u) = & 4(e^{u/2} + e^{-u/2} - 2) - \sum_{n \leq e^u} \frac{\Lambda(n)}{\sqrt{n}}(u - \log n) + \frac{u}{2}(\psi_{\Gamma}(1/4) - \log \pi) \\ & + \frac{1}{4} \left[C_{1/4} - e^{-u/2} \Phi_{\text{L}}(e^{-2u}, 2, 1/4) \right], \quad g_{\text{ar}}(u) = -\Psi_{\text{ar}}(u). \end{aligned} \quad (\text{E.1})$$

Extend both functions evenly to $\mathbb{R}$; $g_{\text{ar}}(0) = 0$. The prime sum is finite at each argument. The notation g_{ar} denotes an arithmetic scalar function, not the Shadow metric g , a source configuration, or a constitutive law.

Proposition E.1 (Laplace and logarithmic-moment normalization). *With the normalization of F and $M = -F'/F$ used in this paper, for $r > 1/2$,*

$$M(-r^2) = \frac{r}{2} \int_0^\infty \Psi_{\text{ar}}(u) e^{-ru} du = -\frac{r}{2} \int_0^\infty g_{\text{ar}}(u) e^{-ru} du. \quad (\text{E.2})$$

In particular, this identity is unconditional in the absolutely convergent prime half-plane.

Proof. Set $s = r + 1/2$. The transform of the elementary term in Equation (E.1) is

$$4 \left(\frac{1}{r - 1/2} + \frac{1}{r + 1/2} - \frac{2}{r} \right) = \frac{1}{r^2} \left(\frac{1}{s - 1} + \frac{1}{s} \right).$$

Each prime-power summand has transform

$$\int_{\log n}^\infty \frac{\Lambda(n)}{\sqrt{n}}(u - \log n) e^{-ru} du = \frac{\Lambda(n) n^{-s}}{r^2}.$$

Nonnegativity permits Tonelli's theorem before the minus sign is restored, and $s > 1$ makes the resulting series absolutely convergent. For the final term set $b = 1/4$ and expand

$$C_{1/4} - e^{-u/2} \Phi_{\text{L}}(e^{-2u}, 2, 1/4) = \sum_{k \geq 0} \frac{1 - e^{-2(k+b)u}}{(k+b)^2}.$$

Its Laplace transform, including the factor $1/4$, is

$$\frac{1}{4r} \sum_{k \geq 0} \frac{1}{(k+b)(k+b+r/2)} = \frac{\psi_{\Gamma}(b+r/2) - \psi_{\Gamma}(b)}{2r^2}.$$

Adding the linear term gives the gamma and $\log \pi$ completion factors. Equation (E.1) therefore has transform $r^{-2} \xi'(s)/\xi(s)$, recovering Suzuki's transform formula in the present variables [29, Theorem 1.1]. Finally, differentiating $F(-r^2) = \xi(1/2+r)/\xi(1/2)$ gives $2rM(-r^2) = \xi'(s)/\xi(s)$ and proves the result. $\square$

The same arithmetic function has a different moment sequence,

$$\mu_n = \frac{1}{4} \int_0^\infty e^{-u/2} \Psi_{\text{ar}}(u) u^n du, \quad \mu_0 = \frac{\xi'(1)}{\xi(1)} = M(-1/4), \quad m_0 = M(0). \quad (\text{E.3})$$

These integrals and their absolute convergence are established in Suzuki's treatment [29, Theorem 1.1 and Equation (1.13)]. The equality at $r = 1/2$ follows by dominated convergence using its bound

$\Psi_{\text{ar}}(u) = O(e^{u/2 - c\sqrt{u}})$, for some $c > 0$. Thus μ_n and m_n are not interchangeable Hankel sequences. Section 13.6 already isolates the resulting central-value discrepancy.

Define, independently of any positivity claim,

$$\mathcal{Q}_\xi(z) = i \frac{\xi'(1/2 - iz)}{\xi(1/2 - iz)}. \quad (\text{E.4})$$

The functional equation and the chain rule give the meromorphic identity

$$\mathcal{Q}_\xi(z) = -\frac{\Xi'(z)}{\Xi(z)} = 2zM(z^2). \quad (\text{E.5})$$

At the origin $\mathcal{Q}_\xi(z)/(2z)$ has the removable value m_0 . This calculation identifies the precise scalar transform attached to the existing logarithmic moments. It does not assert that an arbitrary self-adjoint model realizes that transform. Multiplying ξ by a nonzero constant, including the factor two difference in the 2026 source's completion convention, changes neither Equation (E.4) nor the normalized F .

E.3 Anchored kernels and a constitutive Gram transfer

Define the continuous Hermitian kernel

$$\mathcal{G}(t, u) = g_{\text{ar}}(t - u) - g_{\text{ar}}(t) - g_{\text{ar}}(-u) + g_{\text{ar}}(0). \quad (\text{E.6})$$

Positive kernel means positive semidefiniteness of every finite matrix $(\mathcal{G}(t_i, t_j))$, not nonnegativity of individual entries. Suzuki's screw-function criterion states that this global kernel positivity is equivalent to RH [29, Theorem 1.2]. In the conjugate-linear-first convention of this manuscript, put

$$\langle f, h \rangle_{\mathcal{G}, a} = \int_{-a}^a \int_{-a}^a \overline{f(t)} \mathcal{G}(t, u) h(u) \, du \, dt.$$

For a smooth compactly supported function v , v' has zero integral. Consequently the anchoring terms disappear from the form evaluated on derivatives. With $\tilde{v}(x) = \overline{v(-x)}$, Suzuki's differentiation identity becomes

$$\langle v'_1, v'_2 \rangle_{\mathcal{G}, a} = W(v_2 * \tilde{v}_1), \quad v_1, v_2 \in C_c^\infty(-a, a), \quad (\text{E.7})$$

where W is the Weil functional in that source's convention [29, Proposition 3.1]. The reversal of the two arguments on the right preserves conjugate linearity in v_1 ; it is not an alternative positivity criterion.

Assumption E.2 (Source-to-kernel identification). For each finite window $[-a, a]$, the declared source/readout realization supplies vectors $j_{\sigma, a}(t) \in \text{Dom}(L_\sigma)$ such that

$$\langle L_\sigma j_{\sigma, a}(t), L_\sigma j_{\sigma, a}(u) \rangle = \mathcal{G}(t, u), \quad t, u \in [-a, a]. \quad (\text{E.8})$$

The source norm, derivative domain, and assignment $j_{\sigma, a}$ are defined without assuming RH. Equivalent selectors transport the vectors by the input comparison unitary and the readout derivative by Equation (8.4).

Theorem E.3 (Continuous-kernel Gram transfer). *Assumption E.2 implies RH and hence all moment and determinant equivalences already established in this manuscript. It is sufficient that, for every finite set $T = \{t_1, \dots, t_d\} \subset \mathbb{R}$, there exist positive Gram matrices $C_{r, T}$ with*

$$\max_{i, j} |(C_{r, T})_{ij} - \mathcal{G}(t_i, t_j)| \longrightarrow 0 \quad (r \rightarrow \infty). \quad (\text{E.9})$$

No common Hilbert space for these finite approximants is needed for the positivity conclusion.

Proof. For the exact identification, choose a window containing T and coefficients $c_i \in \mathbb{C}$. The quadratic form of its kernel matrix equals

$$\sum_{i,j} \overline{c_i} c_j \mathcal{G}(t_i, t_j) = \left\| \sum_i c_i L_{\sigma} j_{\sigma,a}(t_i) \right\|^2 \geq 0.$$

In the approximate case, if the maximum entry error is $\epsilon_{r,T}$, its quadratic contribution has magnitude at most

$$\epsilon_{r,T} \left(\sum_i |c_i| \right)^2 \leq d \epsilon_{r,T} \sum_i |c_i|^2.$$

For each fixed finite T , passage to the limit therefore gives a positive semidefinite kernel matrix. Global kernel positivity follows because T was arbitrary, and Suzuki's criterion gives RH. The existing moment and determinant theorems then apply. The argument proves a transfer implication, not the existence of the approximating Gram matrices. Finally, the selector statement follows exactly as in Section 8: the input comparison cancels and the output unitary preserves the norm. $\square$

Equation (E.9) can be restricted to point sets in a fixed dense subset of $\mathbb{R}$. Indeed, continuity of $\mathcal{G}$ allows each arbitrary finite set to be approximated pointwise by such sets, and the finite positive semidefinite cone is closed. This does not reduce the theorem to one finite grid. All finite subsets of the dense set, including arbitrarily large windows and arbitrarily close points, remain necessary. The order of quantifiers is the kernel counterpart of the all-degree limit in Section 12.1.

A stronger global source realization may require compatible window restriction maps, common domains, and an operator limit. Those are separate requirements, just as the common-space trace-norm result in Section 12.2 is stronger than finite moment transfer. Constructing vectors by first assuming positivity of $\mathcal{G}$ would be a conditional Hilbert-space representation, not the independent source construction in Assumption E.2.

E.4 Three spectral constructions and their distinct hypotheses

Connes and van Suijlekom start with a real distribution on $[0, L]$ and its induced quadratic form on trigonometric polynomials. Their precise theorem requires a lower-bounded operator essentially self-adjoint on that core, with a simple isolated lowest eigenvalue and an eigenfunction invariant under $x \mapsto L - x$. The Fourier transform of this extremizer has only real zeros. The one-sided distribution prescription is essential: its symmetrized difference kernel alone can lose boundary information [30, Theorem 6.1 and Remark 4.3]. The extremal and domain hypotheses supplement kernel structure rather than replace it by pointwise positivity, consistently with the counterexample in Section 5.3.

Connes, Consani, and Moscovici place finite arithmetic information in a truncated Weil form and derive a rank-one perturbation of the scaling operator. Their main theorem assumes simplicity of the lowest truncated eigenvalue and evenness of its eigenvector. Self-adjointness uses a quotient equipped with the shifted quadratic-form metric, not merely the original L^2 norm. Their regularized determinant construction is therefore a comparator to the arithmetic realization problem, whereas the limiting identification remains conjectural [31, Theorem 1.1 and Sections 7–8]. Connes separately identifies the need for the appropriate extremal properties and for a sufficiently accurate link between the explicit approximants and the actual extremizers [32, Section 6.6].

Suzuki's September 2026 construction instead uses a self-adjoint extension of $i d/dx$ in a Hilbert completion built from a shifted localized Weil form. Its characteristic functions have real zeros

without assuming the preceding ground-state simplicity or parity conditions. The shift, metric, domains, and normalization nevertheless remain part of the construction [33, Theorem 1.5]. The analytic distinction relevant here is illustrated by writing

$$T_a = A_a - \lambda(a)I > 0, \quad \langle v, A_a v \rangle = \|v\|_{T_a}^2 + \lambda(a)\|v\|_2^2. \quad (\text{E.10})$$

When $\lambda(a) < 0$, positivity of T_a cannot establish positivity of A_a . This is a general algebraic fact, not a defect in a construction that explicitly retains its shift. A source interpretation must derive or declare the metric and shift before comparison and must not hide their dependence in a presumed positive arithmetic norm.

The source identity in Assumption E.2 is a different condition from self-adjointness in any of these arithmetic norms. It asks for an independently specified constitutive readout whose Gram values reproduce the unshifted arithmetic kernel. The external constructions motivate calculable candidates; the identification with a source-derived norm remains to be supplied [6]. Likewise, a finite auxiliary interval or a first-order differential expression does not fall under the ordinary compact-elliptic full-spectrum exclusion merely by sharing some notation. Domains, the Hilbert metric, nonlocal conditions, the spectral variable, and the limiting process all matter.

E.5 The nonlocal characteristic limit is not the determinant F

For a lower-bounded localized operator A_a , Suzuki chooses $\lambda(a) < \inf \text{spec } A_a$. In the corresponding completion, the minimal first-order operator has deficiency indices $(1, 1)$. For equal-norm deficiency vectors $v_\pm$ at $\pm i$, its extension characteristic function is [33, Equation (1.11)]

$$W_a(\theta; z) = (z - i)\widehat{v}_+(z) + e^{i\theta}(z + i)\widehat{v}_-(z). \quad (\text{E.11})$$

In version 3 (23 September 2026), Corollary 1.6, Equation (1.12), on page 7 of [33], the sufficient condition requires choices of $\lambda(a) < \inf \text{spec } A_a$, $\theta(a) \in [0, 2\pi)$, and $\phi(a, \cdot)$ for all sufficiently large a , with convergence as $a \rightarrow \infty$ uniformly on every compact subset of $\mathbb{H}_+ = \{z \in \mathbb{C} : \Im z > 0\}$:

$$e^{\phi(a, z)} W_a(\theta(a); z) \longrightarrow \mathcal{R}_\xi(z) := \frac{\xi(1/2 - iz)}{\xi(1/2 - iz) + \xi'(1/2 - iz)}, \quad (\text{E.12})$$

Here $\phi(a, \cdot)$ is analytic on $\mathbb{H}_+$, so the exponential factor is nowhere zero there; the dependence of W_a on $\lambda(a)$ is suppressed in the notation. The convergence is ordinary locally uniform convergence to a finite holomorphic limit on $\mathbb{H}_+$, not convergence on its boundary or on the whole plane. This is a conditional assertion, not an established limit [33, Corollary 1.6, Equation (1.12)]. It must not be replaced by a purported equality to $F(z^2)$.

Proposition E.4 (Characteristic-ratio normalization and multiplicity). *As meromorphic identities,*

$$\mathcal{R}_\xi(z) = \frac{1}{1 - i\mathcal{Q}_\xi(z)} = \frac{1}{1 - 2izM(z^2)}, \quad M(z^2) = \frac{1 - \mathcal{R}_\xi(z)^{-1}}{2iz}. \quad (\text{E.13})$$

The apparent singularity of the second expression at $z = 0$ is removable. If $s_0 = 1/2 - iz_0$ is a zero of ξ of multiplicity $d \geq 1$, then

$$\mathcal{R}_\xi(z) = -\frac{i}{d}(z - z_0) + O((z - z_0)^2). \quad (\text{E.14})$$

Thus the ratio has a simple zero at each distinct arithmetic zero; the original multiplicity survives in its derivative, not in the order of that zero.

Proof. Equation (E.5) and $\xi'/\xi = -i\mathcal{Q}_\xi$ at $1/2 - iz$ give the first two equalities. Rearrangement gives the inverse relation. Near the origin, $\mathcal{Q}_\xi(z) = 2m_0z + O(z^3)$, hence

$$\mathcal{R}_\xi(z) = 1 + 2im_0z + O(z^2),$$

which establishes removability and recovers $m_0 = \mathcal{R}'_\xi(0)/(2i)$. At a zero s_0 , write $\xi(s) = (s - s_0)^d h(s)$, with $h(s_0) \neq 0$. Then

$$\frac{\xi(s)}{\xi(s) + \xi'(s)} = \frac{(s - s_0)h(s)}{dh(s) + (s - s_0)(h(s) + h'(s))} = \frac{s - s_0}{d} + O((s - s_0)^2).$$

Since $s - s_0 = -i(z - z_0)$, Equation (E.14) follows. The proof also accounts for the cancellation of the common zero of numerator and denominator when $d > 1$. $\square$

This calculation has two consequences for a comparison with the present determinant programme. First, $\mathcal{R}_\xi$ is a meromorphic transform of the logarithmic derivative, not an even entire determinant in the squared variable. Its poles must also be tracked. Second, a simple spectrum for a characteristic function approaching this ratio does not impose simplicity on the zeros of ξ : Equation (E.14) already reduces every zero order to one. Conversely, the exact limiting function contains more information than the set of its zeros, because its derivative retains d . These conclusions follow directly from the displayed identities, without an assumption about the location or simplicity of any zeta zero.

E.6 Locally uniform limits and signed-spectrum bookkeeping

The analytic convergence topology is decisive. The following standard Hurwitz consequence is recorded with the normalization needed to connect an extremal construction to the existing moment tower; the proof is included to display what finite spectral agreement cannot substitute for [26, 30].

Proposition E.5 (Real-zero limits and transfer to logarithmic moments). *Suppose E_r are even entire functions, $E_r(0) = 1$, with only real zeros, and*

$$E_r(z) \longrightarrow \frac{\Xi(z)}{\Xi(0)} \quad \text{locally uniformly on } \mathbb{C}. \quad (\text{E.15})$$

Then RH holds. Writing $D_r(w)$ for the entire function defined by $E_r(z) = D_r(z^2)$, one also has $D_r \rightarrow F$ locally uniformly and

$$[w^n](-D'_r(w)/D_r(w)) \longrightarrow m_n \quad \text{for every fixed } n \geq 0. \quad (\text{E.16})$$

A Fredholm determinant interpretation of D_r requires additional trace-ideal and normalization hypotheses.

Proof. If Ξ had a nonreal zero, choose a closed disc around it disjoint from $\mathbb{R}$ and with no zero on its boundary. The limiting function is not identically zero, since its value at the origin is one. The positive boundary minimum and locally uniform convergence allow Rouché's theorem to force a zero of E_r in the disc for large r , a contradiction. Hence every zero of Ξ is real. For $|w| \leq R$, either square root lies in $|z| \leq \sqrt{R}$, and evenness removes its choice. This gives local uniform convergence of D_r to F . Choose a circle about zero on and inside which F does not vanish. Uniform convergence bounds D_r away from zero there for large r , and Cauchy's estimates give derivative convergence. Applying the coefficient integral to the logarithmic derivatives proves Equation (E.16). $\square$

The proposition deliberately distinguishes analytic convergence from trace-norm convergence. The sequence $D_r(w) = \exp(cw/r)$, with fixed $c \in \mathbb{R} \setminus \{0\}$, tends locally uniformly to 1 and has no zeros; it is nevertheless not a nonconstant positive trace-class Fredholm determinant. More generally, a zero-free exponential factor can survive a poorly normalized entire-function construction. The determinant equality must therefore be established using the conditions of Section 7 or the operator estimates of Section 12.2. Proposition E.5 is a sufficient entire-plane statement; it is not substituted for the different upper-half-plane ratio limit in Equation (E.12). The latter uses the version-3 condition [33, Corollary 1.6, Equation (1.12)]; the two domains of convergence must remain distinct.

Proposition E.6 (Signed spectrum versus reciprocal-square target). *Suppose a self-adjoint operator T has purely discrete spectrum, no zero mode, and eigenvalues $\pm\gamma_j$, where each $\gamma_j > 0$ occurs with the same multiplicity in the positive and negative sectors and $\sum_j \gamma_j^{-2} < \infty$. Put $P_+ = \mathbf{1}_{(0,\infty)}(T)$ and let $K_+ = (T|_{P_+\mathcal{H}})^{-2}$. Then*

$$\mathrm{Tr}(T^{-2})^{n+1} = 2 \mathrm{Tr} K_+^{n+1}, \quad \det(\mathrm{I} - wT^{-2}) = \det(\mathrm{I} - wK_+)^2. \quad (\text{E.17})$$

If the positive spectral sector realizes the arithmetic ordinates, then $\mathrm{Tr} K_+^{n+1} = m_n$. Dividing T^{-2} by two does not repair the doubled tower: its $(n+1)$ st trace is $2^{-n}m_n$.

Proof. On the paired eigenspaces T^{-2} has two copies of every γ_j^{-2} . Summability justifies the trace sums and Fredholm products. Multiplication by $1/2$ changes a power of degree $n+1$ by 2^{-n-1} , giving the final expression after the two copies are counted. $\square$

The condition of paired spectrum is explicit; it is not automatically satisfied by every extension phase in Equation (E.11). Nor can a null mode be removed solely to make an inverse exist. The domain, a source-justified restriction if needed, and a spectral pairing must be established before applying Proposition E.6. This is the first-order counterpart of the square-and-inverse normalization in Section 3.

E.7 Finite-window arithmetic and quantitative rejection

On $[-a, a]^2$, all arguments of g_{ar} in Equation (E.6) lie in $[-2a, 2a]$. Thus a complete prime-power list through any integer $P \geq e^{2a}$ contains the entire arithmetic contribution for this window. There is no infinite prime-tail approximation in that finite kernel. This follows directly from Equation (E.1); it does not remove the need to enlarge the window for global positivity.

For a smaller P , let g_P omit terms with $n > P$ but retain the exact completion terms. Monotonicity of each omitted positive ramp gives

$$0 \leq g_{\mathrm{ar}}(u) - g_P(u) \leq \delta_{P,a} := \sum_{P < n \leq e^{2a}} \frac{\Lambda(n)}{\sqrt{n}} (2a - \log n), \quad |u| \leq 2a. \quad (\text{E.18})$$

For $P \geq e^{2a}$ the sum is empty. Exact integer enumeration fixes which prime powers occur; logarithms, exponentials, and special functions still require controlled evaluation. The bound is not a claim of computational acceleration over established sieves.

Lemma E.7 (Kernel and matrix error propagation). *Let g_1, g_2 be continuous Hermitian functions with $g_1(0) = g_2(0) = 0$ and $\sup_{|u| \leq 2a} |g_1(u) - g_2(u)| \leq \delta$. Write $\mathcal{G}_1, \mathcal{G}_2$ for their anchored kernels. Then*

$$\sup_{t, u \in [-a, a]} |\mathcal{G}_1(t, u) - \mathcal{G}_2(t, u)| \leq 3\delta, \quad \|\mathcal{G}_1[a] - \mathcal{G}_2[a]\|_{2 \rightarrow 2} \leq 6a\delta. \quad (\text{E.19})$$

For a d -point kernel matrix, the spectral norm error is at most $3d\delta$. An additional maximum entry error η in the matrix increases this bound to $d(3\delta + \eta)$.

Proof. There are three varying terms in the anchored difference; the common value at zero cancels. The integral kernel bound has row and column integrals at most $2a \cdot 3\delta$, so the Schur estimate proves the operator bound. For matrices, the quadratic form bound used in Theorem E.3 gives the spectral norm estimate. The triangle inequality accounts for the additional entry error. $\square$

In particular, let a Hermitian numerical matrix $\hat{G}$ approximate the arithmetic matrix G with a *proved* error $\|G - \hat{G}\| \leq b$. A verified negative Rayleigh value below $-b$ contradicts positivity of G ; a verified discrepancy between a proposed positive source Gram matrix and G beyond the combined error budget rejects that proposed identification. A positive finite matrix establishes only its finite test. Numerical agreement at higher precision, by itself, is not the proved enclosure b . If the origin value is also approximated instead of fixed exactly, the kernel bound must include its fourth error contribution.

The separate implementation `screw_kernel_diagnostics.py` evaluates the finite-window arithmetic function, tests the two meromorphic normalization identities at complex arguments, checks the central/quarter-shift distinction, and performs exact rational tests of signed-spectrum counting, nonlinear aggregation, and entrywise-to-quadratic error propagation. Its sample windows use only complete finite prime-power lists. Outputs at 40 and 60 decimal digits are numerical diagnostics, not interval certificates and not an evaluation of a source-derived kernel. No zeta-zero list is supplied to the calculation. The legacy diagnostics and their filenames remain separate.

For the four-point sets $\{-a, -a/2, a/2, a\}$, Table 5 records the smallest eigenvalue of the unweighted arithmetic kernel matrix. The two working precisions agree at every displayed digit. These are finite matrix evaluations rather than lower bounds for the continuum operator; positivity at these points is not promoted to positivity on an interval.

Table 5: Finite arithmetic-kernel diagnostics. The cutoffs cover all prime powers needed for the stated windows; eigenvalues are non-certified numerical evaluations.

Window radius a	Complete cutoff P	Smallest sampled eigenvalue
1/2	3	0.0400010208048755
1	8	0.0363429964746402
2	55	0.0341769970295997

E.8 The source-compatible completion problem

The preceding identities furnish a concrete arithmetic intermediate object, not another source primitive. At a fixed admissible configuration, a proposed realization must specify the source/readout norms and L_σ , then construct either $j_{\sigma,a}(t)$ or a corresponding insertion of test functions. The kernel equality must be verified before invoking positive Gram geometry. Conservative cochain assembly and interior elimination supply a possible route to the source side, but the shared boundary functional and the arithmetic insertion remain the separate requirements already identified in Sections 2 and 8 [20, 21, 6].

Three limits must not be conflated: refinement $r \rightarrow \infty$ at fixed window, window expansion $a \rightarrow \infty$, and analytic convergence in the complex spectral parameter. Theorem E.3 permits the first

limit for every fixed finite set. Proposition E.5 requires locally uniform complex convergence. A proposed operator determinant requires trace control as well. Neither a cutoff a nor a refinement label r is automatically the gauge label of a physical foliation. The source ontology, auxiliary representation coordinates, and operational clock retain the distinctions of Section 2 [4].

Table 6: Arithmetic intermediates and the remaining identifications.

Object	Established role	Additional requirement
$g_{\text{ar}}, \mathcal{G}$	Explicit arithmetic function and anchored kernel; Equation (E.2) connects the transform to M .	Independent source-to-kernel Gram identification on every window.
$\mathcal{R}_\xi$	Meromorphic transform of M ; zero multiplicity is encoded in the derivative as in Proposition E.4.	The exact complex-function limit, not substitution by F or by its zero set.
Extremal Fourier transforms	Real zeros under the applicable distributional and extremal hypotheses.	Normalized analytic convergence to the specified arithmetic target.
Positive shifted metric	A legitimate Hilbert norm when its operator is strictly positive.	Control of the shift and proof of the unshifted arithmetic positivity or exact identity.
Paired first-order spectrum	Squaring doubles the reciprocal-square traces.	Positive-sector selection justified before comparison; all trace factors retained.
Finite arithmetic matrix	Exact prime-power support with finite evaluation and approximation errors.	Error enclosures for certificates and an all-window limit for RH.

The resulting extension strengthens the constructive options without weakening the existing exclusions. A local Morse potential, a shifted Weil norm, a boundary response, a screw kernel, and a characteristic ratio are not equivalent objects. Their comparison becomes meaningful precisely through the displayed maps and normalizations. The new sufficient transfer statements do not furnish their own source witnesses: constructing those witnesses, or the required nonlocal analytic limit, is the remaining arithmetic realization problem.

Data, code, and computational assistance

No experimental dataset is used. The numerical inputs are the defining special functions, exact constants, and parameter choices printed in the manuscript. The source package contains `main.tex`, `diagnostics.py`, `morse_diagnostics.py`, and machine-readable outputs. It also contains `robin_shift_diagnostics.py`, which checks the added coefficient algebra and reproduces the Robin normalization and large-order comparisons.

The package additionally contains `scale_endpoint_diagnostics.py`, `shifted_tower_diagnostics.py`, `completed_tower_diagnostics.py`, `kinetic_rate_diagnostics.py`, and

`sesquilinear_diagnostics.py`, with their machine-readable outputs. There are ten scientific diagnostic scripts in total, with two separate verification drivers. The driver `verify_release.py` reruns the nine scripts in its explicit manifest, including `boundary_trace_diagnostics.py` and `sesquilinear_diagnostics.py`, in 13 fresh-directory executions. It checks the regenerated JSON files against the packaged outputs and the printed values in Tables 1–2 and Equations (13.20)–(13.21); it does not invoke `screw_kernel_diagnostics.py`.

The script `sesquilinear_diagnostics.py` performs 35 exact Gaussian-rational checks of the conjugate-linear first-slot convention, conjugate symmetry, scalar and phase laws, real–imaginary decomposition, the trace–Gram identity, and polynomial-adjoint conjugation, including negative controls with the first-slot conjugation deliberately omitted. These manufactured checks verify the conventions in Sections 6.1, 8.1, and 12.1; they do not construct the arithmetic insertion map.

Language-model assistance was used in developing the mathematical argument, drafting, preparing code, and checking LaTeX consistency. This disclosure is not a claim of independent peer verification; responsibility for final verification and submission rests with the named author.

The geometric source papers are cited as manuscripts at their stated fourth-amendment dates; no repository identifier is invented for them. The additional script `boundary_trace_diagnostics.py` and its JSON output use exact rational arithmetic and are separate from the unchanged floating-point xi and Morse diagnostics. The amended source package also includes a preservation report and a source manifest distinguishing source-derived statements, proved conditional implications, and unexecuted arithmetic closure.

The tenth scientific script, `screw_kernel_diagnostics.py`, is accompanied by separate 40- and 60-digit outputs and is rerun by `verify_screw_release.py`, not by `verify_release.py`. This second driver checks both outputs and the three printed rows of Table 5, giving six printed-value comparisons; its 56 distinct exact integer/rational checks are counted once across the two precisions. It does not replace any earlier diagnostic. Its computations distinguish exact integer/rational checks from arbitrary-precision evaluations; no numerical output is treated as an interval certificate. The versioned external works supporting Appendix E are listed individually in the bibliography.

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References

- [1] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsberichte der Berliner Akademie (1859), 671–680. Original text and translations: <https://www.maths.tcd.ie/pub/HistMath/People/Riemann/Zeta/>.
- [2] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986.
- [3] Clay Mathematics Institute, *Riemann Hypothesis*, Millennium Prize Problems, <https://www.claymath.org/millennium/riemann-hypothesis/>, accessed 5 September 2026.

- [4] H. Arellano-Peña, *Timeless Counterspace & Shadow Gravity—A Unified Framework: 4D-Counterduction, Foundational Consistency, and Cartographic Inquiry*, Preprints, version 2, 11 August 2026, not peer reviewed. <https://doi.org/10.20944/preprints202511.1472.v2>.
- [5] H. Arellano, *Source–Shadow Geometry of TCGS–SEQUENTION: Gravito-Capillary Foams, a Retrocausal Non-Local Foliation, and a Disciplined Cartographic Program for Cluster Offsets, Cosmic Anisotropy, and Biological Invariants*, v3.0 integrated, closure-audited preprint, June 2026. <https://www.researchgate.net/publication/397427092>.
- [6] H. Arellano-Peña, *Discrete Records Alone Do Not Determine Gauge Ontology: Connes Rigidity, Lattice Continuum Limits, and Source–Shadow Reconstruction in TCGS–SEQUENTION*, preprint, 5 August 2026. <https://doi.org/10.13140/RG.2.2.33830.89921>.
- [7] M. V. Berry and J. P. Keating, The Riemann zeros and eigenvalue asymptotics, *SIAM Review* **41** (1999), 236–266. <https://doi.org/10.1137/S0036144598347497>.
- [8] X.-J. Li, The positivity of a sequence of numbers and the Riemann hypothesis, *Journal of Number Theory* **65** (1997), 325–333.
- [9] D. A. Cardon and S. A. Roberts, An equivalence for the Riemann Hypothesis in terms of orthogonal polynomials, *Journal of Approximation Theory* **138** (2006), 54–64. <https://doi.org/10.1016/j.jat.2005.09.016>.
- [10] A. Voros, *Zeta Functions over Zeros of Zeta Functions*, Lecture Notes of the Unione Matematica Italiana, vol. 8, Springer, 2010. <https://doi.org/10.1007/978-3-642-05203-3>.
- [11] J. Grommer, Ganze transzendente Funktionen mit lauter reellen Nullstellen, *Journal für die reine und angewandte Mathematik* **144** (1914), 114–166. <https://doi.org/10.1515/crll.1914.144.114>. Archival record: <https://eudml.org/doc/149420>.
- [12] K. Schmüdgen, *The Moment Problem*, Graduate Texts in Mathematics, vol. 277, Springer, Cham, 2017. <https://doi.org/10.1007/978-3-319-64546-9>.
- [13] B. Simon, *Trace Ideals and Their Applications*, 2nd ed., Mathematical Surveys and Monographs, vol. 120, American Mathematical Society, Providence, 2005.
- [14] M. Reed and B. Simon, *Methods of Modern Mathematical Physics, I: Functional Analysis*, revised and enlarged ed., Academic Press, New York, 1980.
- [15] L. Hörmander, The spectral function of an elliptic operator, *Acta Mathematica* **121** (1968), 193–218. <https://doi.org/10.1007/BF02391913>.
- [16] J. C. Lagarias, The Schrödinger operator with Morse potential on the right half-line, *Communications in Number Theory and Physics* **3** (2009), 323–361. <https://doi.org/10.4310/CNTP.2009.v3.n2.a3>. Preprint: <https://arxiv.org/abs/0712.3238>.
- [17] National Institute of Standards and Technology, *Digital Library of Mathematical Functions*, Chapters 5, 10, 13, and 25, <https://dlmf.nist.gov/>, accessed 5 September 2026.
- [18] E. Bombieri, *Problems of the Millennium: The Riemann Hypothesis*, Clay Mathematics Institute, official problem description. <https://www.claymath.org/wp-content/uploads/2022/05/riemann.pdf>.

- [19] The mpmath development team, *mpmath: A Python Library for Arbitrary-Precision Floating-Point Arithmetic*, software version 1.3.0 used in the computations. <https://mpmath.org/>.
- [20] H. Arellano-Peña, *Conservative Discrete Source-to-Readout Maps in Four Dimensions: Accountability Conditions, Median-Dual Geometry, Singular-Orbit Topology, and Boundary Completion, with a TCGS-SEQUENTION Application*, manuscript, fourth amendment, 28 August 2026, CC BY 4.0.
- [21] H. Arellano-Peña, *Reciprocal Fractal-Foam Geometry of Counterspace: Core Geometry, Conservative Discretization, Domain-Merger Benchmark, and Falsification Guide*, manuscript, fourth amendment, 28 August 2026, CC BY 4.0.
- [22] D. N. Arnold, R. S. Falk, and R. Winther, Finite element exterior calculus: from Hodge theory to numerical stability, *Bulletin of the American Mathematical Society* **47** (2010), 281–354. <https://doi.org/10.1090/S0273-0979-10-01278-4>. Preprint: <https://arxiv.org/abs/0906.4325>.
- [23] F. Dörfler and F. Bullo, *Kron Reduction of Graphs with Applications to Electrical Networks*, preprint, 2011. <https://arxiv.org/abs/1102.2950>.
- [24] A. Girouard and I. Polterovich, *Spectral Geometry of the Steklov Problem*, preprint, 2014. <https://arxiv.org/abs/1411.6567>.
- [25] F. Bornemann, On the numerical evaluation of Fredholm determinants, *Mathematics of Computation* **79** (2010), 871–915. <https://doi.org/10.1090/S0025-5718-09-02280-7>. Preprint: <https://arxiv.org/abs/0804.2543>.
- [26] B. Ja. Levin, *Distribution of Zeros of Entire Functions*, revised ed., Translations of Mathematical Monographs, vol. 5, American Mathematical Society, Providence, 1980.
- [27] Á. Baricz and F. Štampach, The Hurwitz-type theorem for the regular Coulomb wave function via Hankel determinants, *Linear Algebra and its Applications* **548** (2018), 259–272. <https://doi.org/10.1016/j.laa.2018.03.012>. Preprint: <https://arxiv.org/abs/1708.07729v3>.
- [28] T. Craven and G. Csordas, Iterated Laguerre and Turán inequalities, *Journal of Inequalities in Pure and Applied Mathematics* **3** (2002), no. 3, Article 39, 14 pp. <https://eudml.org/doc/122436>.
- [29] M. Suzuki, Aspects of the screw function corresponding to the Riemann zeta-function, *Journal of the London Mathematical Society* **108** (2023), no. 4, 1448–1487. <https://doi.org/10.1112/jlms.12785>. Preprint: <https://arxiv.org/abs/2206.03682v4>.
- [30] A. Connes and W. D. van Suijlekom, Quadratic Forms, Real Zeros and Echoes of the Spectral Action, *Communications in Mathematical Physics* **406** (2025), Article 312. <https://doi.org/10.1007/s00220-025-05493-1>. Preprint: <https://arxiv.org/abs/2511.23257v1>.
- [31] A. Connes, C. Consani, and H. Moscovici, *Zeta Spectral Triples*, preprint, version 1, 27 November 2025. <https://arxiv.org/abs/2511.22755v1>.
- [32] A. Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, preprint, version 1, 3 February 2026. <https://arxiv.org/abs/2602.04022v1>.

- [33] M. Suzuki, *Weil's quadratic form via the screw function*, preprint, version 3, revised 23 September 2026. <https://arxiv.org/abs/2606.09096v3>.
- [34] B. W. López Subia, Fórmula para hallar la cifra de números primos menores que una cantidad dada, *Revista Ciencia, Tecnología e Innovación* **18** (2020), no. 22, 125–148. Published online 23 February 2021. <https://doi.org/10.56469/rcti.v18i22.396>.
- [35] B. W. López Subia, La demostración rigurosa de la fórmula $\pi(x)$ y la criba de Beimar Wilfredo López Subia, *Revista Ingenium* **1**, no. 2, 67–78; issue dated November 2021. Journal landing page records publication on 1 November 2024. <https://revistas.usfx.bo/index.php/ing/article/view/1529>. Full text: <https://revistas.usfx.bo/index.php/ing/article/download/1529/1198/4192>.